Hitting time and enumeration of loose Hamilton cycles in random hypergraphs

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Abstract

The random hypergraph process starts with an empty hypergraph on n vertices and adds a uniformly chosen missing edge at each step. For every fixed $r \geq 4$ with $(r-1) \mid n$, we prove that, with high probability, an r-uniform loose Hamilton cycle appears when the last isolated vertex disappears. We also estimate the number of loose Hamilton cycles at this time up to a factor $\exp\{O_r(n/\log n)\}$, with the reference value expressed in terms of the actual hitting time. The proof adapts entropy, conditioned sampling, and random edge-deletion arguments from Kahn's work on perfect matchings. A biased distribution of intersection and private vertices and two injective cycle modifications extend these arguments to higher uniformity.

1 Introduction

A *loose Hamilton cycle* in an r-graph is a spanning subhypergraph whose edges admit a cyclic ordering in which consecutive edges intersect in exactly one vertex, these intersection vertices are distinct, and nonconsecutive edges are disjoint. Each edge contains two intersection vertices and $r-2$ private vertices. Thus a loose Hamilton cycle on n vertices has $n/(r-1)$ edges, and $(r-1) \mid n$ is necessary.

Fix an integer $r \geq 4$. Let $(\mathcal{H}_t)_{t=0}^{\binom{n}{r}}$ be the random r-uniform hypergraph process on $[n] \coloneqq \{1, \dots, n\}$. More precisely, set $\mathcal{H}_0 = \varnothing$ and, conditional on $\mathcal{H}_t$, choose an edge uniformly from $\binom{[n]}{r} \setminus \mathcal{H}_t$ and add it to obtain $\mathcal{H}_{t+1}$. We define two hitting times in this process: τ_1 is the time when the last isolated vertex disappears, and τ_{LHC} is the time when a loose Hamilton cycle first appears. That is,

$$\tau_1 \coloneqq \min\{t \colon \delta_1(\mathcal{H}_t) \geq 1\}, \quad \text{and} \quad \tau_{\mathrm{LHC}} \coloneqq \min\{t \colon \mathcal{H}_t \text{ contains a loose Hamilton cycle}\},$$

where δ_1 denotes minimum vertex degree. Observe that a loose Hamilton cycle has no isolated vertex, so deterministically $\tau_{\mathrm{LHC}} \geq \tau_1$.

We prove the following hitting-time statement.

Theorem 1.1. *For every fixed $r \geq 4$, as n tends to infinity through multiples of $r-1$, $\mathbb{P}[\tau_{\mathrm{LHC}} = \tau_1]$ tends to 1.*

For an r-graph $\mathcal{G}$, let $Z(\mathcal{G})$ denote its number of loose Hamilton cycles. The proof also estimates this number at the hitting time, retaining the actual value of τ_1 in the reference expression.

Theorem 1.2. *For every fixed $r \geq 4$, there is a constant $C_r > 0$ such that, as n tends to infinity through multiples of $r-1$, with probability $1 - o(1)$,*

$$\left| \log Z(\mathcal{H}_{\tau_1}) - \left(\frac{n}{r-1} \log \frac{r(r-1)\tau_1}{n} - n \right) \right| \leq \frac{C_r n}{\log n}. \tag{1.1}$$

In particular, with high probability,

$$Z(\mathcal{H}_{\tau_1}) = \exp\{o(n)\} \left(\frac{(r-1)\log n}{e^{r-1}} \right)^{n/(r-1)}. \tag{1.2}$$

The hitting-time problem for loose Hamilton cycles was raised by Kühn and Osthus [23, Section 5.2] and revisited by Frieze and Pérez-Giménez [15, Section 5]. Theorem 1.1 gives the conclusion for every fixed $r \geq 4$. Frieze and Pérez-Giménez [15, Theorem 1] proved that, for every fixed $\gamma > 0$ and $r \geq 3$, a uniform

random m-edge r-graph contains a loose Hamilton cycle whp when $m \geq (1+\gamma)n \log n/r$ and $(r-1) \mid n$. This determines the threshold up to a factor of $1 + o(1)$.

We briefly review earlier results and the methods related to the proof. In the binomial r-uniform hypergraph model, every r-set is present independently with probability p. Frieze [13] and Dudek and Frieze [7] proved loose Hamiltonicity at the threshold order $\log n/n^{r-1}$ under additional divisibility assumptions. Dudek, Frieze, Loh, and Speiss [8] removed these additional assumptions, and Ferber [11] gave a coupling proof. The threshold order also follows from Frankston, Kahn, Narayanan, and Park [12, Theorem 5.1].

In the random regular r-uniform hypergraph model, Dudek, Frieze, Ruciński, and Šileikis [9] obtained loose Hamiltonicity for d-regular hypergraphs with $d = \omega(\log n)$ and $d = o(n^{1/2})$, for fixed $r \geq 3$ and the necessary divisibility conditions. Their proof embeds a uniform random hypergraph with a prescribed number of edges in the random regular hypergraph. Their later work [10] extends this embedding to $d = \omega(\log n)$ and $d = o(n^{r-1})$. For fixed uniformity and fixed degree, Altman, Greenhill, Isaev, and Ramadurai [1] determined the degree threshold for loose Hamiltonicity.

The proof below encodes mixed Hamilton cycles as perfect matchings. For perfect matchings in random uniform hypergraphs, early threshold bounds were obtained by Schmidt and Shamir [25], Frieze and Janson [14], and Kim [22]. Johansson, Kahn, and Vu [19] determined the threshold order, and Kahn [21] obtained the asymptotically sharp threshold. Kahn [20, Theorems 1.3 and 1.4] also proved the corresponding hitting-time theorem and a counting lower bound at the disappearance of the last isolated vertex. The proof here adapts his entropy, conditioned sampling, and random edge-deletion arguments. Cuckler and Kahn [5, 6] used edge marginals and entropy to count perfect matchings and Hamilton cycles in ordinary graphs. Here the entropy inequality is Kahn's hypergraph matching bound; the cycle modifications are needed to preserve a single cyclic component and every prescribed pair.

We first reveal the edges incident to vertices of very small degree at τ_1. For each such vertex, choose an incident edge and designate two other vertices of that edge as intersection vertices. The exceptional vertex then belongs to the private block of size $r-2$. Delete this block and replace the chosen edge by the prescribed pair joining its intersection vertices. We also reveal every edge meeting a deleted block. With high probability the exceptional vertices are sufficiently separated for these operations to be disjoint. Conditional on the revealed edges, the hypergraph induced by the remaining vertices is uniform subject to its edge count and the resulting degree lower bounds. A mixed Hamilton cycle through the prescribed pairs expands to a loose Hamilton cycle in $\mathcal{H}_{\tau_1}$.

We count mixed cycles by randomly deleting edges from the complete hypergraph down to this conditioned core. Assuming a lower bound on the number of remaining cycles supplies an entropy lower bound. After encoding cycles as perfect matchings, Kahn's inequality controls most directed edge marginals. A comparison of forward and reverse sampling extends this estimate to most choices of a deleted private block and an added directed pair, even when the corresponding edge is absent. Injective migrations then convert this average control into a uniform edge-marginal bound. Together with the martingale estimate, this rules out a first failure of the counting lower bound.

Two features distinguish higher uniformity. The proportion of intersection vertices is $1/(r-1)$, so the entropy argument compares their indicator distribution with a biased Bernoulli product law. Also, a completion deletes a private block of $r-2$ vertices. We change these vertices one at a time, then change the two intersection vertices; the resulting comparisons involve only a bounded number of deleted vertices for fixed r.

The basic definitions and enumeration appear in Section 2. The matching and mixed-cycle entropy estimates are proved in Sections 3 and 4, followed by the degree-constrained model in Section 5. The reduction and deletion argument in Sections 6 and 7 leave a uniform edge-marginal bound to prove. The conditional laws, edge resampling, and migrations in Sections 8 to 10 establish this bound. We finish with the count at the hitting time in Section 11.

2 Preliminaries

We first introduce the hypergraph notation and the mixed cycles used in the reduction. All hypergraphs are simple. For a finite set V, write $\binom{V}{r}$ for the family of its r-element subsets. We usually identify a hypergraph $\mathcal{G}$ with its edge set, and hence use $|\mathcal{G}|$ for its number of edges. For $S \subseteq V(\mathcal{G})$, let $\mathcal{G} - S$ be

the hypergraph on $V(\mathcal{G}) \setminus S$ consisting of the edges disjoint from S. For a single vertex v, write $\mathcal{G} - v$ for $\mathcal{G} - \{v\}$.

For a hypergraph $\mathcal{G}$, the vertex degree $d_{\mathcal{G}}(v)$ counts its edges containing v, and the codegree $d_{\mathcal{G}}(u, v)$ counts its edges containing distinct vertices u, v. Write $\delta_1(\mathcal{G})$ for the minimum vertex degree, $\Delta_1(\mathcal{G})$ for the maximum vertex degree, and $\Delta_2(\mathcal{G})$ for the maximum codegree. Its average degree is $d(\mathcal{G}) \coloneqq r|\mathcal{G}|/|V(\mathcal{G})|$. The link $L_{\mathcal{G}}(v)$ consists of the $(r-1)$-sets $A \subseteq V(\mathcal{G}) \setminus \{v\}$ for which $A \cup \{v\} \in \mathcal{G}$. For disjoint sets $A, B \subseteq V(\mathcal{G})$, let

$$\mathcal{G}[A, B] \coloneqq \{e \in \mathcal{G} \colon |e \cap A| = 2 \text{ and } e \setminus A \subseteq B\}.$$

When $B = V(\mathcal{G}) \setminus A$, we write $\mathcal{G}[A, \overline{A}]$.

For the edge-distribution estimates, define

$$\theta_r \coloneqq \binom{r}{2} \frac{1}{(r-1)^2} \left(\frac{r-2}{r-1}\right)^{r-2}.$$

This is the asymptotic fraction of all r-sets having exactly two vertices in a set of size $n/(r-1)$.

A *Berge path* of length t consists of distinct vertices $v_0, \dots, v_t$ and distinct edges $e_1, \dots, e_t$ such that $\{v_{i-1}, v_i\} \subseteq e_i$ for every $i \in [t]$.

Definition 2.1. *Let V be an n-vertex set, let $\mathcal{G}$ be an r-graph on V, and let M be a prescribed matching of s pairs on V. We use the following terminology and notation.*

(i) *A* $(\mathcal{G}, M)$-respecting mixed Hamilton cycle *is a hypergraph on V consisting of all pairs of M and r-edges of $\mathcal{G}$. Its edges admit a cyclic ordering in which consecutive edges intersect in exactly one vertex, these intersection vertices are distinct, and nonconsecutive edges are disjoint. The vertices used are exactly V.*

(ii) *A vertex is* M-saturated *if it belongs to a pair of M, and* M-unsaturated *otherwise.*

(iii) *Write $\Phi(\mathcal{G}, M)$ for the number of these cycles. For a mixed cycle or path C, let $E_r(C)$ be its set of r-edges. The other edges of C are precisely the pairs in M.*

Each r-edge has two intersection vertices and a *private block* of $r-2$ vertices. These private blocks are pairwise disjoint and avoid every intersection vertex. We sometimes call the r-edges ordinary edges when distinguishing them from prescribed pairs.

Fact 2.2. *If a mixed Hamilton cycle contains k ordinary edges and s prescribed pairs, then the following statements hold.*

(i) *Every M-saturated vertex belongs to one ordinary edge and one prescribed pair.*

(ii) *There are $k+s$ intersection vertices, of which $2s$ are M-saturated, and $(r-2)k$ private vertices. Consequently,*

$$n = (r-1)k + s.$$

For a nonempty M, choose a pair as the *root pair* and fix its direction. Every mixed Hamilton cycle then has a unique orientation agreeing with that direction.

The next lemma gives the number of mixed Hamilton cycles when the host hypergraph is complete.

Lemma 2.3. *Let M be a matching of $s \geq 1$ pairs on an n-vertex set V. Suppose $k = (n-s)/(r-1)$ is an integer with $k \geq \max\{3, s\}$, and write $d = r-2$ within this lemma. Then*

$$\Phi\left(\tbinom{V}{r}, M\right) = 2^{s-1}(k-1)!\binom{n-2s}{k-s}\frac{(dk)!}{(d!)^k}. \tag{2.1}$$

If $k \geq 2s$, then requiring every ordinary edge to contain at most one M-saturated vertex multiplies this number by

$$\frac{(k-s)!(k-s-1)!}{(k-1)!(k-2s)!} \geq 1 - \frac{s(s-1)}{k-1}. \tag{2.2}$$

Thus, if $s = o(\sqrt{n})$, this restriction preserves a $1 - o(1)$ fraction of the mixed Hamilton cycles.

Proof. We count cycles by arranging k blocks in a cyclic order, where each prescribed pair is a block and each M-unsaturated intersection vertex is a singleton block. Choose the $k-s$ singleton blocks from the $n-2s$ unsaturated vertices. Fix a direction on the root pair. The blocks have $(k-1)!$ cyclic orders, and the remaining prescribed pairs have 2^{s-1} choices of direction. Assign the dk private vertices to the k positions between successive blocks, using an unordered d-set at each position. There are $(dk)!/(d!)^k$ such assignments. These choices uniquely determine the cycle, and its root orientation recovers them, proving (2.1).

The restriction in the second assertion is equivalent to having no two prescribed-pair blocks consecutive. Among cyclic arrangements of k blocks, the fraction with this property is

$$\frac{\frac{k}{k-s}\binom{k-s}{s}}{\binom{k}{s}} = \frac{(k-s)!(k-s-1)!}{(k-1)!(k-2s)!}.$$

For each prescribed-pair block, its successor is another such block with probability $(s-1)/(k-1)$. A union bound over the s blocks gives the lower bound in (2.2). The assignments of private vertices are unaffected. □

We next record the contraction used to count cycles containing a specified ordinary edge. Let G be an r-graph, let M be a nonempty prescribed matching on $V(G)$, and fix a directed root pair in M. For a set $P \subseteq V(G) \setminus V(M)$ of size $r-2$ and distinct $y, z \in V(G) \setminus (P \cup V(M))$, let $\Psi_{G,M}(P; y \to z)$ be the number of $(G-P, M \cup \{yz\})$-respecting mixed Hamilton cycles in which the new prescribed pair is directed from y to z relative to the root direction. This number is defined whether or not $P \cup \{y, z\}$ belongs to G.

Fact 2.4. *Suppose each original mixed cycle has at least four ordinary edges.*

(i) The two possible directions of the new pair partition its cycle family, so

$$\Phi(G-P, M \cup \{yz\}) = \Psi_{G,M}(P; y \to z) + \Psi_{G,M}(P; z \to y). \tag{2.3}$$

(ii) For every $e \in G$ disjoint from $V(M)$,

$$|\{C : C \text{ is } (G,M)\text{-respecting and } e \in E_r(C)\}| = \sum_{\{y,z\} \in \binom{e}{2}} \Phi\big(G - (e \setminus \{y,z\}), M \cup \{yz\}\big). \tag{2.4}$$

Indeed, contracting the private block of e replaces e by its two intersection vertices as a prescribed pair. The resulting cycle has $n-(r-2)$ vertices, $s+1$ prescribed pairs, and one fewer ordinary edge. The added pair records the two intersection vertices of the contracted r-edge.

Fact 2.5. *Let $G = \binom{V}{r}$ be the complete r-graph on n vertices, and let M contain $s \geq 1$ prescribed pairs. Suppose $k = (n-s)/(r-1)$ is an integer with $k \geq 4$ and $k - s \geq 2$. For P, y, z as above,*

$$\frac{\Psi_{G,M}(P; y \to z)}{\Phi(G,M)} = \frac{(r-2)!(k-s)(k-s-1)}{(k-1)(n-2s)_r}. \tag{2.5}$$

Here $(a)_b = a(a-1)\cdots(a-b+1)$. If $s = o(n)$, this ratio is $(1+o(1))/((r-1)^2 d(G))$.

Proof. When the host hypergraph is complete, reversing the new pair block gives a bijection between its two relative orientation classes. Hence the numerator is half the count in (2.1) with parameters $(n-(r-2), s+1, k-1)$. Dividing by the original count and cancelling the factorials gives (2.5). □

The entropy argument will identify $\Phi(G,M)/((r-1)^2 d(G))$ as the corresponding scale for sparse hosts.

We now introduce the probability notation used below. All logarithms are natural.

- For real-valued random variables X, Y, write $X \preceq Y$ if $\mathbb{P}[X \geq t] \leq \mathbb{P}[Y \geq t]$ for every real t; in other words, Y *stochastically dominates* X.
- An event holds *with high probability* (whp) if its probability tends to 1 as the number of vertices tends to infinity.
- For a nonempty finite set S, write $\mathrm{Unif}(S)$ for its uniform distribution. Thus, if $X \sim \mathrm{Unif}(S)$, then $\mathbb{P}[X = x] = 1/|S|$ for every $x \in S$.

- If T is a uniformly chosen t-element subset of a finite set U and $A \subseteq U$ is fixed, then $|T \cap A|$ has a *hypergeometric distribution*.
- Write Poisson(λ) for the Poisson distribution of mean λ.
- For a finite-valued random vector Z under a law $\mathbb{P}$, a value z is $\mathbb{P}$-*admissible* if $\mathbb{P}[Z = z] > 0$.

Whenever values are specified in a conditional law, they are assumed to be admissible for that law. We use the convention $\log 0 = -\infty$ in inequalities involving cycle numbers. Constants may depend on the fixed uniformity r and the other constants specified in the statement, but not on vertex labels or a matching fixed before sampling.

The following tail bounds will be used for degrees and random edge sets; see Janson, Łuczak, and Ruciński [18, Chapter 2].

Lemma 2.6. *Let X be binomial or hypergeometric with mean $\lambda > 0$, and define $\varphi(a) := 1 - a + a \log a$ for $a > 0$. Then the following statements hold.*

(i) *For $0 < a < 1$, $\mathbb{P}[X \le a\lambda] \le \exp\{-\lambda\varphi(a)\}$.*

(ii) *For $t \ge \lambda$, $\mathbb{P}[X \ge t] \le (e\lambda/t)^t$.*

(iii) *For $0 < \delta \le 1$, $\mathbb{P}[|X - \lambda| \ge \delta\lambda] \le 2\exp\{-\delta^2\lambda/3\}$.*

For a nonnegative integer-valued $X \preceq \mathrm{Poisson}(\lambda)$ and a positive integer t, we also use $\mathbb{P}[X \ge t] \le \lambda^t/t!$. Increasing events in a product probability space are positively associated by Harris's inequality [16].

For a finite-valued random variable Z with probabilities p_z, its entropy is $\mathbb{H}[Z] := -\sum_z p_z \log p_z$. For probability distributions p, q on the same finite set, their relative entropy is

$$D_{\mathrm{KL}}[p\|q] := \sum_z p_z \log \frac{p_z}{q_z},$$

with $0 \log 0 = 0$ and a positive term with $q_z = 0$ interpreted as $+\infty$. We also write $\mathbb{H}[p]$ for the entropy of a finite probability distribution p. Conditional entropy is the average of the entropies under the conditional laws:

$$\mathbb{H}[Z \mid W] := \sum_w \mathbb{P}[W = w]\mathbb{H}[Z \mid W = w].$$

We use the chain rule $\mathbb{H}[Z, W] = \mathbb{H}[W] + \mathbb{H}[Z \mid W]$, nonnegativity of relative entropy, and the log-sum inequality; see Cover and Thomas [4].

Lemma 2.7. *For probability distributions p, q on a finite set, $\sum_z |p_z - q_z| \le \sqrt{2D_{\mathrm{KL}}[p\|q]}$.*

We also use Shearer's inequality [3]: if a collection of coordinate subsets covers each coordinate of a finite random vector at least b times, then the sum of the entropies of the corresponding restrictions is at least b times the entropy of the whole vector. These inequalities apply to each admissible conditional distribution as well.

3 Kahn's matching entropy inequality and its stability

Kahn proved the following inequality, which compares the entropy of a perfect matching with the entropies of the matching edges containing individual vertices. We use it to derive the edge-marginal bounds in Proposition 3.2, and then apply these bounds to mixed Hamilton cycles in Section 4.

Theorem 3.1 ([21, Theorem 4.2])**.** *Let $r \ge 3$, let $r \mid n$, and let $\mathbf{F}$ have any probability distribution on the perfect matchings of an n-vertex r-graph $\mathcal{G}$. For $e \in \mathcal{G}$, put $p_e := \mathbb{P}[e \in \mathbf{F}]$ and*

$$\gamma_e := \mathbb{P}\big[\mathbf{F}(u) = \mathbf{F}(v) \text{ for some distinct } u, v \in e\big],$$

where $\mathbf{F}(u)$ is the matching edge containing u. Then

$$\mathbb{H}[\mathbf{F}] \le \frac{1}{r}\sum_{u \in V(\mathcal{G})} \mathbb{H}[\mathbf{F}(u)] - \frac{(r-1)n}{r} + O_r\left(\sum_{u \in V(\mathcal{G})}\sum_{e \ni u} p_e \gamma_e^{1/(r-1)}\right) + O_r(\log n).$$

Since every perfect matching has n/r edges, the vector $(rp_e/n)_{e\in\mathcal{G}}$ is a probability distribution on $\mathcal{G}$. The next result bounds its relative entropy with respect to the uniform distribution.

Proposition 3.2. *Fix $r \geq 3$ and $C \geq 1$. Let $\mathcal{G}$ be an n-vertex r-graph with $r \mid n$, $d(\mathcal{G}) \to \infty$, $\Delta_1(\mathcal{G}) \leq Cd(\mathcal{G})$, and $\Delta_2(\mathcal{G}) = o(d(\mathcal{G}))$. Let $\mathbf{F}$ have any probability distribution on its perfect matchings, and put $p_e \coloneqq \mathbb{P}[e \in \mathbf{F}]$. Then*

$$\mathbb{H}[\mathbf{F}] + \frac{n}{2r} D_{\mathrm{KL}}\left[\left(\frac{rp_e}{n}\right)_{e\in\mathcal{G}} \middle\| \mathrm{Unif}(\mathcal{G})\right] \leq \frac{n}{r}\log d(\mathcal{G}) - \frac{(r-1)n}{r} + O_{r,C}\left(\frac{n}{\log(d(\mathcal{G})/\Delta_2(\mathcal{G}))} + \log n\right). \quad (3.1)$$

Consequently, if $\mathbb{H}[\mathbf{F}] \geq (n/r)\log d(\mathcal{G}) - (r-1)n/r - \varepsilon n$ for some $\varepsilon \geq 0$, then

$$\sum_{e\in\mathcal{G}} \left| p_e - \frac{1}{d(\mathcal{G})} \right| = O_{r,C}\left(n\sqrt{\varepsilon + \frac{1}{\log(d(\mathcal{G})/\Delta_2(\mathcal{G}))} + \frac{\log n}{n}}\right).$$

Proof. We bound the collision term in Theorem 3.1 by a small multiple of the loss in the local matching entropies. That loss is the relative entropy in the statement, up to an additive term of order n.

Claim 3.3. *Let D be an integer with $D \geq \Delta_1(\mathcal{G})$ and $\Delta_2(\mathcal{G}) = o(D)$. Then*

$$\sum_{u\in V(\mathcal{G})} \sum_{e\ni u} p_e \gamma_e^{1/(r-1)} \leq O_r\left(n\left(\frac{\Delta_2(\mathcal{G})}{D}\right)^{1/(2(r-1))}\right) + O_r\left(\frac{n\log D - \sum_{u\in V(\mathcal{G})}\mathbb{H}[\mathbf{F}(u)] + n}{\log(D/\Delta_2(\mathcal{G}))}\right).$$

Proof. A perfect matching contains $(n/r)\binom{r}{2}$ pairs of vertices lying in a common matching edge. Every edge whose vertices fail to lie in distinct matching edges contains one of these pairs. Each pair lies in at most $\Delta_2(\mathcal{G})$ host edges. Counting the r incidences of each such host edge and taking expectations gives

$$\sum_{u\in V(\mathcal{G})} \sum_{e\ni u} \gamma_e \leq n\binom{r}{2}\Delta_2(\mathcal{G}).$$

At each vertex u, consider the distribution $(p_e)_{e\ni u}$ of $\mathbf{F}(u)$, and append entries of weight 0 until it has D entries. Regard each original entry as a position (u,e) with value γ_e, and assign value 0 to every appended position. The weights p_e/n on the original positions, together with the appended weights 0, form a probability distribution on these nD positions. Its relative entropy with respect to the uniform distribution on the positions is

$$\sum_{u\in V(\mathcal{G})} \sum_{e\ni u} \frac{p_e}{n} \log \frac{p_e/n}{1/(nD)} = \log D - \frac{1}{n}\sum_{u\in V(\mathcal{G})} \mathbb{H}[\mathbf{F}(u)].$$

Let B consist of the original positions (u,e) with γ_e greater than $\sqrt{\binom{r}{2}\Delta_2(\mathcal{G})/D}$. The preceding collision bound gives

$$|B| \leq nD\sqrt{\frac{\binom{r}{2}\Delta_2(\mathcal{G})}{D}}.$$

If B is empty, then every γ_e is at most this threshold and the first term in the claim suffices. Otherwise, the log-sum inequality on B and its complement, together with the bound $\log 2$ for the entropy of their two total weights, gives

$$\left(\sum_{(u,e)\in B} p_e\right)\log\frac{nD}{|B|} \leq n\log D - \sum_{u\in V(\mathcal{G})}\mathbb{H}[\mathbf{F}(u)] + n\log 2.$$

The bound on $|B|$ gives $\log(nD/|B|) \geq \frac{1}{2}\log(D/(\binom{r}{2}\Delta_2(\mathcal{G})))$. On the complement of B, the factor $\gamma_e^{1/(r-1)}$ is at most $(\binom{r}{2}\Delta_2(\mathcal{G})/D)^{1/(2(r-1))}$. Summing separately over the two sets proves the claim. □

We now take $D = \lceil Cd(\mathcal{G}) \rceil$. Since each edge has r incidences and $\sum_e p_e = n/r$, we have the identity

$$\frac{1}{r} \sum_{u \in V(\mathcal{G})} \mathbb{H}[\mathbf{F}(u)] = \frac{n}{r} \log d(\mathcal{G}) - \frac{n}{r} D_{\mathrm{KL}} \left[\left(\frac{rp_e}{n} \right)_{e \in \mathcal{G}} \middle\| \mathrm{Unif}(\mathcal{G}) \right]. \tag{3.2}$$

In particular,

$$n \log D - \sum_{u \in V(\mathcal{G})} \mathbb{H}[\mathbf{F}(u)] = n \log \frac{D}{d(\mathcal{G})} + n D_{\mathrm{KL}} \left[\left(\frac{rp_e}{n} \right)_{e \in \mathcal{G}} \middle\| \mathrm{Unif}(\mathcal{G}) \right].$$

The first term is $O_C(n)$. Also, for fixed r, $x^{1/(2(r-1))} = O_r(1/\log(1/x))$ as $x \to 0$. Substituting Claim 3.3 and (3.2) into Theorem 3.1 therefore gives

$$\begin{aligned} \mathbb{H}[\mathbf{F}] + \frac{n}{r} D_{\mathrm{KL}} \left[\left(\frac{rp_e}{n} \right)_{e \in \mathcal{G}} \middle\| \mathrm{Unif}(\mathcal{G}) \right] &\le \frac{n}{r} \log d(\mathcal{G}) - \frac{(r-1)n}{r} \\ &\quad + O_{r,C} \left(\frac{n}{\log(d(\mathcal{G})/\Delta_2(\mathcal{G}))} \left(1 + D_{\mathrm{KL}} \left[\left(\frac{rp_e}{n} \right)_{e \in \mathcal{G}} \middle\| \mathrm{Unif}(\mathcal{G}) \right] \right) + \log n \right). \end{aligned}$$

The coefficient of the relative entropy in the error is $o(n)$, so it is at most $n/(2r)$ for sufficiently large n. Moving this term to the left proves (3.1). For the final conclusion, compare with the assumed entropy lower bound and apply Lemma 2.7, multiplied by n/r. The uniform reference marginal is $n/(r|\mathcal{G}|) = 1/d(\mathcal{G})$. □

4 Entropy stability for mixed Hamilton cycles

We now encode mixed Hamilton cycles as perfect matchings and apply Proposition 3.2. There are two sources of variation. A vertex outside the prescribed pairs may be an intersection vertex or a private vertex, and a prescribed pair may have either direction relative to the root. We first condition on these choices. The entropy lower bound controls both the conditional matching marginals and the distribution of the choices themselves.

Throughout this section, fix a root pair in the nonempty prescribed matching M and an arbitrary direction on it. Orient every mixed Hamilton cycle to agree with this direction. For a private block P of size $r-2$ and distinct vertices u, v outside P, write $(P; u \to v)$ for the oriented edge that has private block P and goes from u to v. Such an oriented edge is allowed when $P \cup \{u, v\} \in \mathcal{G}$. Let $\mathbf{C}$ have a probability distribution on the $(\mathcal{G}, M)$-respecting mixed Hamilton cycles. Relative to this specified cycle distribution, we write

$$p(P; u \to v) := \mathbb{P}[\mathbf{C} \text{ uses } P \cup \{u, v\} \text{ with private block } P \text{ and direction } u \to v].$$

Theorem 4.1. *Fix $r \ge 4$ and $C \ge 1$. Let $\mathcal{G}$ be an m-edge r-graph on n vertices with $d(\mathcal{G}) \to \infty$. Let M be a nonempty matching of $s = o(n)$ prescribed pairs, with $(r-1) \mid n - s$. Suppose that $\Delta_1(\mathcal{G}) \le Cd(\mathcal{G})$, $\Delta_2(\mathcal{G}) = o(d(\mathcal{G}))$, and, for every $A \supseteq V(M)$ with $|A| = (n-s)/(r-1) + s$,*

$$\left| |\mathcal{G}[A, \overline{A}]| - \theta_r m \right| \le \delta n d(\mathcal{G}), \quad 0 \le \delta = o(1).$$

Suppose the random mixed Hamilton cycle $\mathbf{C}$ satisfies

$$\mathbb{H}[\mathbf{C}] \ge \frac{n-s}{r-1} \log((r-1)d(\mathcal{G})) - (n-s) - \xi n, \quad 0 \le \xi = o(1).$$

Then

$$\begin{aligned} &\sum_{\substack{e \in \mathcal{G} \\ e \cap V(M) = \varnothing}} \sum_{\substack{u,v \in e \\ u \ne v}} \left| p(e \setminus \{u, v\}; u \to v) - \frac{1}{(r-1)^2 d(\mathcal{G})} \right| \\ &\qquad = O_{r,C} \left(n \sqrt{\xi + \delta + \frac{s}{n} + \frac{1}{\log(d(\mathcal{G})/\Delta_2(\mathcal{G}))}} \right). \end{aligned} \tag{4.1}$$

Proof. Put $k \coloneqq (n-s)/(r-1)$ within this proof. Let $\mathbf{J}$ be the set of M-unsaturated intersection vertices of $\mathbf{C}$, and let $\mathbf{\Theta}$ record the directions of the pairs in M other than the root pair. Then $|\mathbf{J}| = k - s$.

For a set $J_0 \subseteq V(\mathcal{G}) \setminus V(M)$ and directions θ on the nonroot pairs, call (J_0, θ) feasible if they arise from a $(\mathcal{G}, M)$-respecting mixed Hamilton cycle. For each feasible pair, construct $\widetilde{\mathcal{G}}_{J_0,\theta}$ as follows.

(i) Replace every vertex of J_0 by an incoming copy and an outgoing copy. If a prescribed pair goes from a to b, keep only the incoming copy at a and the outgoing copy at b. Keep every vertex outside $J_0 \cup V(M)$ unchanged.

(ii) For each $P \cup \{u, v\} \in \mathcal{G}$ with an outgoing copy at u, an incoming copy at v, and P consisting entirely of unchanged private vertices, include the edge $\{u^{\mathrm{out}}, v^{\mathrm{in}}\} \cup P$.

Once (J_0, θ) is fixed, this auxiliary hypergraph is fixed. Conditional on $\mathbf{J} = J_0$ and $\mathbf{\Theta} = \theta$, the ordinary edges of $\mathbf{C}$ give a random perfect matching $\mathbf{F}$ of $\widetilde{\mathcal{G}}_{J_0,\theta}$. Its distribution is induced by the cycle distribution and need not be uniform on all perfect matchings. The matching and the prescribed pairs recover the directed cycle. Thus the chain rule gives

$$\mathbb{H}[\mathbf{C}] = \mathbb{H}[\mathbf{J}, \mathbf{\Theta}] + \mathbb{H}[\mathbf{F} \mid \mathbf{J}, \mathbf{\Theta}]. \tag{4.2}$$

In conditional sums below, $\widetilde{\mathcal{G}}$ denotes the auxiliary hypergraph determined by the realized values of $(\mathbf{J}, \mathbf{\Theta})$; the outer expectation averages over these values. Put

$$\lambda_0 \coloneqq \left(\frac{r-2}{r-1}\right)^{r-2} d(\mathcal{G}), \quad \varepsilon \coloneqq \xi + \delta + \frac{s}{n} + \frac{1}{\log(d(\mathcal{G})/\Delta_2(\mathcal{G}))}.$$

Claim 4.2. *For every feasible (J_0, θ), the following statements hold.*

(i) The auxiliary hypergraph has k incoming vertices, k outgoing vertices, and $(r-2)k$ private vertices, hence rk vertices in total.

(ii) Its maximum degree is at most $2\Delta_1(\mathcal{G})$ and its maximum codegree is at most $2\Delta_2(\mathcal{G})$.

(iii) Its average degree satisfies $d(\widetilde{\mathcal{G}}_{J_0,\theta}) = \lambda_0(1 + O_{r,C}(\delta + s/n))$.

Proof. There are $k - s$ unsaturated intersection vertices and s copies of each direction supplied by the prescribed-pair vertices. The remaining $n - 2s - (k - s) = (r-2)k$ vertices are private, proving (i).

An original edge has at most two lifts after the intersection and private vertices are fixed. Two copies of the same original vertex lie in no auxiliary edge. Every auxiliary edge through vertices with distinct original labels projects to a host edge through those labels. This proves (ii).

Apply the edge-distribution hypothesis to $A = J_0 \cup V(M)$. Every edge in $\mathcal{G}[A, \overline{A}]$ disjoint from $V(M)$ has two lifts. Edges meeting $V(M)$ contribute an error of at most $2s\Delta_1(\mathcal{G}) = O_{r,C}(sd(\mathcal{G}))$. Consequently,

$$|\widetilde{\mathcal{G}}_{J_0,\theta}| = 2\theta_r m + O_{r,C}((\delta n + s)d(\mathcal{G})).$$

Divide by k to obtain the auxiliary average degree. The definition of θ_r and $n = (r-1)k + s$ give

$$\frac{2\theta_r m}{k} = \left(\frac{r-2}{r-1}\right)^{r-2} \frac{n}{n-s} d(\mathcal{G}) = \lambda_0(1 + O_r(s/n)),$$

proving (iii). □

The next claim controls both the entropy of $(\mathbf{J}, \mathbf{\Theta})$ and the conditional matching marginals.

Claim 4.3. *We have*

$$\log \binom{n-2s}{k-s} + (s-1)\log 2 - \mathbb{H}[\mathbf{J}, \mathbf{\Theta}] + \frac{k}{2}\mathbb{E}\left[D_{\mathrm{KL}}\left[\left(\frac{\mathbb{P}[f \in \mathbf{F} \mid \mathbf{J}, \mathbf{\Theta}]}{k}\right)_{f \in \widetilde{\mathcal{G}}} \middle\| \mathrm{Unif}(\widetilde{\mathcal{G}})\right]\right] = O_{r,C}(n\varepsilon).$$

Moreover,

$$\mathbb{E}\left[\sum_{f\in\widetilde{\mathcal{G}}}\left|\mathbb{P}[f\in\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]-\frac{1}{\lambda_0}\right|\right]=O_{r,C}(n\sqrt{\varepsilon}). \tag{4.3}$$

Proof. By Claim 4.2, every conditional auxiliary host satisfies the hypotheses of Proposition 3.2, with constants depending only on r and C. Its error is $O_{r,C}(n\varepsilon)$. Here the $O_r(\log n)$ term is absorbed because $1\le\Delta_2(\mathcal{G})$ and $d(\mathcal{G})\le\binom{n-1}{r-1}$, so $\log n=O_r(n/\log(d(\mathcal{G})/\Delta_2(\mathcal{G})))$ for large n. Apply (3.1) under each conditional matching law, average, and use (4.2). This gives

$$\mathbb{H}[\mathbf{C}]+\frac{k}{2}\mathbb{E}\left[D_{\mathrm{KL}}\left[\left(\frac{\mathbb{P}[f\in\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]}{k}\right)_{f\in\widetilde{\mathcal{G}}}\middle\|\,\mathrm{Unif}(\widetilde{\mathcal{G}})\right]\right]$$
$$\le\mathbb{H}[\mathbf{J},\mathbf{\Theta}]+k\log\lambda_0-(r-1)k+O_{r,C}(n\varepsilon).$$

There are at most $\binom{n-2s}{k-s}2^{s-1}$ possible values of $(\mathbf{J},\mathbf{\Theta})$. Stirling's formula gives

$$\log\binom{n-2s}{k-s}+(s-1)\log 2+k\log\lambda_0=k\log((r-1)d(\mathcal{G}))+O_r(s+\log n). \tag{4.4}$$

To see the cancellation of the linear terms, put $h(x)=-x\log x-(1-x)\log(1-x)$ and use

$$(r-1)h\left(\frac{1}{r-1}\right)+(r-2)\log\frac{r-2}{r-1}=\log(r-1).$$

The changes caused by s are $O_r(s)$ since $s=o(n)$ and the limiting proportions lie strictly between 0 and 1.

Subtract the assumed entropy lower bound for $\mathbf{C}$ and use (4.4). Both terms in the first conclusion of the claim are nonnegative, so each is $O_{r,C}(n\varepsilon)$. Pinsker's inequality and Cauchy–Schwarz now give

$$\mathbb{E}\left[\sum_{f\in\widetilde{\mathcal{G}}}\left|\mathbb{P}[f\in\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]-\frac{1}{d(\widetilde{\mathcal{G}})}\right|\right]\le k\sqrt{2\mathbb{E}\left[D_{\mathrm{KL}}\left[\left(\frac{\mathbb{P}[f\in\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]}{k}\right)_{f\in\widetilde{\mathcal{G}}}\middle\|\,\mathrm{Unif}(\widetilde{\mathcal{G}})\right]\right]}=O_{r,C}(n\sqrt{\varepsilon}).$$

Since $|\widetilde{\mathcal{G}}|=kd(\widetilde{\mathcal{G}})$, replacing $1/d(\widetilde{\mathcal{G}})$ by $1/\lambda_0$ costs

$$|\widetilde{\mathcal{G}}|\left|\frac{1}{d(\widetilde{\mathcal{G}})}-\frac{1}{\lambda_0}\right|=k\left|1-\frac{d(\widetilde{\mathcal{G}})}{\lambda_0}\right|=O_{r,C}(n\delta+s).$$

This is absorbed by $n\sqrt{\varepsilon}$, proving (4.3). □

For each allowed oriented edge $(P;u\to v)$ disjoint from $V(M)$, its lift is present precisely when $u,v\in\mathbf{J}$ and $P\cap\mathbf{J}=\varnothing$. Projecting (4.3) and applying the triangle inequality gives

$$\sum_{\substack{e\in\mathcal{G}\\ e\cap V(M)=\varnothing}}\sum_{\substack{u,v\in e\\ u\neq v}}\left|p(e\setminus\{u,v\};u\to v)-\frac{\mathbb{P}[u,v\in\mathbf{J},\ e\setminus\{u,v\}\subseteq V(\mathcal{G})\setminus\mathbf{J}]}{\lambda_0}\right|$$
$$=O_{r,C}(n\sqrt{\varepsilon}). \tag{4.5}$$

We finish by controlling the membership probabilities appearing here.

Claim 4.4. *We have*

$$\sum_{\substack{e\in\mathcal{G}\\ e\cap V(M)=\varnothing}}\sum_{\substack{u,v\in e\\ u\neq v}}\left|\mathbb{P}[u,v\in\mathbf{J},\ e\setminus\{u,v\}\subseteq V(\mathcal{G})\setminus\mathbf{J}]-\frac{1}{(r-1)^2}\left(\frac{r-2}{r-1}\right)^{r-2}\right|$$
$$=O_{r,C}(nd(\mathcal{G})\sqrt{\varepsilon}).$$

Proof. Let ν be the distribution of the indicator vector of $\mathbf{J}$ on the $n-2s$ unsaturated vertices. It has exactly $k-s$ entries equal to 1. The first conclusion of Claim 4.3 and the entropy bound on $\mathbf{\Theta}$ give

$$\log\binom{n-2s}{k-s} - \mathbb{H}[\mathbf{J}] \le \log\binom{n-2s}{k-s} + (s-1)\log 2 - \mathbb{H}[\mathbf{J},\mathbf{\Theta}] = O_{r,C}(n\varepsilon).$$

We compare ν with independent Bernoulli variables of mean $1/(r-1)$. Expanding the cross entropy and applying Stirling's formula gives

$$\begin{aligned} D_{\mathrm{KL}}\left[\nu \,\middle\|\, \mathrm{Bernoulli}\left(\frac{1}{r-1}\right)^{\otimes(n-2s)}\right] &= \log\binom{n-2s}{k-s} - \mathbb{H}[\mathbf{J}] \\ + (n-2s)h\left(\frac{k-s}{n-2s}\right) - \log\binom{n-2s}{k-s} &+ (n-2s)D_{\mathrm{KL}}\left[\mathrm{Bernoulli}\left(\frac{k-s}{n-2s}\right) \,\middle\|\, \mathrm{Bernoulli}\left(\frac{1}{r-1}\right)\right] \\ &= O_{r,C}(n\varepsilon) + O_r(\log n + s^2/n) = O_{r,C}(n\varepsilon). \end{aligned}$$

For the last equality, use $(k-s)/(n-2s) = 1/(r-1) + O_r(s/n)$ and the quadratic relative-entropy bound between Bernoulli laws whose parameters stay away from 0 and 1.

For each $e \in \mathcal{G} - V(M)$, let ν_e be the joint distribution of the membership indicators $(\mathbf{1}_{v\in\mathbf{J}})_{v\in e}$. Apply Shearer's inequality to these coordinate sets, adding singleton sets so that each coordinate occurs exactly $\Delta_1(\mathcal{G})$ times. Cross entropies against the product Bernoulli distribution add over coordinates. Subtracting the entropy-cover inequality and discarding the nonnegative singleton relative entropies therefore gives

$$\begin{aligned} &\sum_{e\in\mathcal{G}-V(M)} D_{\mathrm{KL}}\left[\nu_e \,\middle\|\, \mathrm{Bernoulli}\left(\frac{1}{r-1}\right)^{\otimes r}\right] \\ &\qquad \le \Delta_1(\mathcal{G}) D_{\mathrm{KL}}\left[\nu \,\middle\|\, \mathrm{Bernoulli}\left(\frac{1}{r-1}\right)^{\otimes(n-2s)}\right] = O_{r,C}(nd(\mathcal{G})\varepsilon). \end{aligned}$$

A membership pattern with exactly two intersection vertices determines two oriented edges. By Pinsker's inequality and Cauchy–Schwarz, the sum in the claim is at most

$$2\sqrt{2m\sum_{e\in\mathcal{G}-V(M)} D_{\mathrm{KL}}\left[\nu_e \,\middle\|\, \mathrm{Bernoulli}\left(\frac{1}{r-1}\right)^{\otimes r}\right]} = O_{r,C}(nd(\mathcal{G})\sqrt{\varepsilon}).$$

This proves the claim. □

Divide the bound in Claim 4.4 by λ_0 and combine it with (4.5). The reference probability becomes

$$\frac{1}{\lambda_0(r-1)^2}\left(\frac{r-2}{r-1}\right)^{r-2} = \frac{1}{(r-1)^2 d(\mathcal{G})},$$

which proves (4.1). □

The following consequence supplies the error bound used in the resampling argument.

Corollary 4.5. *Fix $r \ge 4$, $B \ge 0$, and $C \ge 1$. Let $\mathcal{H}$ be an m-edge r-graph on n vertices with $d(\mathcal{H}) \ge (\log n)/C$. Let M be a matching of $1 \le s \le Cn^{1/10}$ prescribed pairs, with $(r-1) \mid n-s$. Suppose that the following bounds hold.*

(i) We have $\Delta_1(\mathcal{H}) \le Cd(\mathcal{H})$ and $\Delta_2(\mathcal{H}) \le Cd(\mathcal{H})/(\log n)^{1/4}$.

(ii) For every $A \supseteq V(M)$ with $|A| = (n-s)/(r-1)+s$, we have $\left||\mathcal{H}[A,\overline{A}]| - \theta_r m\right| \le Cm/(\log n)^{1/8}$.

Let $\mathcal{G} \subseteq \mathcal{H}$ be obtained by removing at most $Cn^{1/10}d(\mathcal{H})$ edges. Suppose a nonempty family $\mathcal{C}$ of $(\mathcal{G},M)$-respecting mixed Hamilton cycles satisfies

$$\log|\mathcal{C}| \ge \frac{n-s}{r-1}\log((r-1)d(\mathcal{H})) - (n-s) - \frac{Bn}{\sqrt{\log n}}.$$

For a uniform cycle in $\mathcal{C}$,

$$\sum_{\substack{e\in\mathcal{G}\\ e\cap V(M)=\varnothing}}\sum_{\substack{u,v\in e\\ u\neq v}}\left|p(e\setminus\{u,v\};u\to v)-\frac{1}{(r-1)^2d(\mathcal{H})}\right|=O_{r,B,C}\left(\frac{n}{\sqrt{\log\log n}}\right). \tag{4.6}$$

For every $t>0$, the number of these oriented edges with relative deviation greater than t is $O_{r,B,C}(m/(t\sqrt{\log\log n}))$, with a constant independent of t.

Proof. The number of removed edges gives $d(\mathcal{G})=d(\mathcal{H})(1+O_{r,C}(n^{-9/10}))$. The edge-distribution bound transfers to $\mathcal{G}$ because both its edge count on any fixed partition and the reference value change by $O_{r,C}(n^{1/10}d(\mathcal{H}))$. The hypotheses of Theorem 4.1 therefore hold with $\delta=O_{r,C}((\log n)^{-1/8})$, $\Delta_2(\mathcal{G})/d(\mathcal{G})=O_{r,C}((\log n)^{-1/4})$, $\xi=O_{r,B,C}((\log n)^{-1/2})$, and $s/n=O_{r,C}(n^{-9/10})$. Substituting these bounds into (4.1) gives an error of order $n/\sqrt{\log\log n}$. Replacing $1/((r-1)^2d(\mathcal{G}))$ by $1/((r-1)^2d(\mathcal{H}))$ costs $O_{r,C}(n^{1/10})$ in the sum, proving (4.6). Every orientation with relative deviation greater than t contributes more than $t/((r-1)^2d(\mathcal{H}))$ to this sum. The final bound follows from $nd(\mathcal{H})=rm$. □

A prescribed direction on the root pair orients every cycle uniquely. A prescribed relative direction on a second pair restricts the family, so its actual cardinality must satisfy the entropy lower bound before Corollary 4.5 is applied. In particular, keeping at least an n^{-A} fraction of a family, for fixed A, loses at most $A\log n$ from the logarithm of its cardinality.

We also need an overlap estimate under weaker hypotheses. The next lemma uses only the coarse entropy lower bound and the maximum-degree bound; it does not require a codegree or edge-distribution estimate.

Lemma 4.6. *Fix $r\geq 4$ and $B,C\geq 1$. Let $\mathcal{C}$ be a nonempty family of mixed Hamilton cycles respecting a fixed nonempty matching of prescribed pairs. Suppose every cycle has $k\asymp_r n$ ordinary edges, the host $\mathcal{G}$ has average degree $d(\mathcal{G})\to\infty$ and maximum degree at most $Cd(\mathcal{G})$, and*

$$\log|\mathcal{C}|\geq k\log d(\mathcal{G})-Bn.$$

For independent uniform cycles $\mathbf{C}_1,\mathbf{C}_2\in\mathcal{C}$,

$$\mathbb{E}[|E_r(\mathbf{C}_1)\cap E_r(\mathbf{C}_2)|]=O_{r,B,C}\left(\frac{n}{\log d(\mathcal{G})}\right). \tag{4.7}$$

Proof. Orient the cycles by a fixed direction on a prescribed pair. Let $\mathbf{J}$ and $\mathbf{\Theta}$ be the intersection set and pair directions induced by $\mathbf{C}_1$, and construct its auxiliary matching $\mathbf{F}$ as above. The number of possible values of $(\mathbf{J},\mathbf{\Theta})$ is $\exp\{O_r(n)\}$, so the chain rule gives

$$\mathbb{H}[\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]=\log|\mathcal{C}|-\mathbb{H}[\mathbf{J},\mathbf{\Theta}]\geq k\log d(\mathcal{G})-O_{r,B}(n).$$

Take $D=\lceil 2Cd(\mathcal{G})\rceil$, which bounds every auxiliary degree. For each admissible value (J_0,θ), Shearer's inequality applied to the auxiliary edge indicators gives

$$r\mathbb{H}[\mathbf{F}\mid\mathbf{J}=J_0,\mathbf{\Theta}=\theta]\leq\sum_{u\in V(\widetilde{\mathcal{G}}_{J_0,\theta})}\mathbb{H}[\mathbf{F}(u)\mid\mathbf{J}=J_0,\mathbf{\Theta}=\theta].$$

Equivalently, averaging over the realized values of $(\mathbf{J},\mathbf{\Theta})$ gives

$$\mathbb{E}\left[\sum_{u\in V(\widetilde{\mathcal{G}})}\left(\log D+\sum_{f\ni u}\mathbb{P}[f\in\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]\log\mathbb{P}[f\in\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]\right)\right]$$
$$\leq rk\log D-r\mathbb{H}[\mathbf{F}\mid\mathbf{J},\mathbf{\Theta}]=O_{r,B,C}(n).$$

For a conditional matching law, write $p_u^{\max}$ for its largest incidence probability at u. Splitting off an outcome of this probability bounds its entropy by $\log 2+(1-p_u^{\max})\log D$. Consequently,

$$\mathbb{E}\left[\sum_{u\in V(\widetilde{\mathcal{G}})}p_u^{\max}\right]=O_{r,B,C}\left(\frac{n}{\log d(\mathcal{G})}\right).$$

For each fixed value (J_0, θ), an original edge has at most two auxiliary versions. Hence

$$\sum_{e\in\mathcal{G}} \mathbb{P}[e \in E_r(\mathbf{C}_1) \mid \mathbf{J} = J_0, \mathbf{\Theta} = \theta]^2 \le 2 \sum_{f\in\widetilde{\mathcal{G}}_{J_0,\theta}} \mathbb{P}[f \in \mathbf{F} \mid \mathbf{J} = J_0, \mathbf{\Theta} = \theta]^2$$
$$= \frac{2}{r} \sum_{u\in V(\widetilde{\mathcal{G}}_{J_0,\theta})} \sum_{f\ni u} \mathbb{P}[f \in \mathbf{F} \mid \mathbf{J} = J_0, \mathbf{\Theta} = \theta]^2 \le \frac{2}{r} \sum_{u\in V(\widetilde{\mathcal{G}}_{J_0,\theta})} p_u^{\max}.$$

Jensen's inequality allows us to average this bound and gives $\sum_{e\in\mathcal{G}} \mathbb{P}[e \in E_r(\mathbf{C}_1)]^2 = O_{r,B,C}(n/\log d(\mathcal{G}))$. By independence of $\mathbf{C}_1$ and $\mathbf{C}_2$, this sum is their expected ordinary-edge overlap, proving (4.7). □

5 Random hypergraphs with degree lower bounds

A key object in the proof is the family of hypergraphs satisfying prescribed degree lower bounds. For a positive integer n, a nonnegative integer m, and a vector $\boldsymbol{\ell} = (\ell_v)_{v\in[n]}$ of nonnegative integers, define

$$\Omega_r(n, m, \boldsymbol{\ell}) := \left\{ \mathcal{G} \subseteq \binom{[n]}{r} : |\mathcal{G}| = m \text{ and } d_{\mathcal{G}}(v) \ge \ell_v \text{ for every } v \in [n] \right\}.$$

Throughout this section, $r \ge 4$ is fixed. We call $(n, m, \boldsymbol{\ell})$

- *admissible* if $\Omega_r(n, m, \boldsymbol{\ell}) \neq \varnothing$;
- *near-critical* if

$$\left| m - \frac{n \log n}{r} \right| \le \frac{3n \log\log n}{r}, \quad \text{and} \quad |\ell_v - \lfloor 0.01 \log n \rfloor| \le 10r \quad \text{for every } v \in [n]. \tag{5.1}$$

We first use a switching to bound the degree of a specified vertex.

Lemma 5.1. *Let $\mathcal{G}$ be uniform in a nonempty $\Omega_r(n, m, \boldsymbol{\ell})$. For every $v \in [n]$,*

$$\max\left\{0, d_{\mathcal{G}}(v) - \max\left\{\ell_v, \left\lceil \frac{rm}{n} \right\rceil\right\}\right\} \preceq \text{Poisson}\left(\frac{rm}{n}\right).$$

Proof. Write A_t for the number of members of $\Omega_r(n, m, \boldsymbol{\ell})$ with degree t at v. If $t > \max\{\ell_v, rm/n\}$, then replace an edge $\{v\} \cup S$ by $\{w\} \cup S$, where $w \notin S \cup \{v\}$ and the new edge is absent. This preserves all degree lower bounds: the degree at v decreases by one, and the degree at w increases by one. There are $t(n-r)$ choices before excluding existing new edges, and at most $r(m-t)$ choices are excluded. The number of legal forward switchings is therefore at least $nt - rm$. A reverse switching has at most $r(m-t+1)$ choices. Double counting gives $(nt-rm)A_t \le r(m-t+1)A_{t-1}$, and hence $A_t \le \frac{rm/n}{t-rm/n} A_{t-1}$.

Let p_q be the probability that the random variable on the left of the claimed domination equals q. For $q \ge 1$, the recurrence gives $p_q \le \frac{rm}{nq} p_{q-1}$. For $q = 1$, the probability p_0 assigned to zero includes the probabilities of all degrees up to $\max\{\ell_v, \lceil rm/n \rceil\}$, so it is at least the probability of the last of these layers. If π_q is the Poisson mass of mean rm/n, then $\pi_q/\pi_{q-1} = rm/(nq)$, and therefore p_q/π_q is nonincreasing. Fix an integer $j \ge 1$. If $p_j \le \pi_j$, then the same monotonicity gives $p_q \le \pi_q$ for all $q \ge j$, so summing proves the upper-tail comparison. Otherwise $p_q \ge \pi_q$ for all $q < j$, and hence

$$\sum_{q\ge j} p_q = 1 - \sum_{q<j} p_q \le 1 - \sum_{q<j} \pi_q = \sum_{q\ge j} \pi_q.$$

This proves stochastic domination. The case $m = 0$ is immediate. □

A degree-preserving switching gives a corresponding bound for sums of codegrees.

Lemma 5.2. *Fix a realizable degree sequence $\boldsymbol{d} = (d_v)_{v\in[n]}$, and choose $\mathcal{G}$ uniformly among its simple r-graph realizations. Let $B \subseteq [n]$ and $y \in [n] \setminus B$. If*

$$\sum_{v\in[n]\setminus B} d_v - r^2\Delta_1(\mathcal{G}) - 2\Delta_1(\mathcal{G})^2 > 0,$$

then

$$\sum_{v\in B} d_{\mathcal{G}}(y, v) \preceq \text{Poisson}\left(\frac{(r-1)d_y \sum_{v\in B} d_v}{\sum_{v\in[n]\setminus B} d_v - r^2\Delta_1(\mathcal{G}) - 2\Delta_1(\mathcal{G})^2} \right).$$

Proof. For every integer $t \geq 0$, let A_t count realizations of $\boldsymbol{d}$ in which the sum on the left equals t. Choose an incidence (e, z) with $y, z \in e$ and $z \in B$, and a disjoint edge f with a distinguished vertex $a \in f \setminus B$. Exchange z and a, provided both new edges are absent. Degrees are preserved and the codegree sum decreases by one.

For each of the t choices of (e, z), start with $\sum_v d_v$ choices of (f, a). Exclude $\sum_{v \in B} d_v$ choices with $a \in B$, at most $r^2 \Delta_1(\mathcal{G})$ choices for which f meets e, and at most $\Delta_1(\mathcal{G})^2$ choices for each of the two possible existing new edges. A reverse move is specified by an edge through y, a distinguished vertex of that edge other than y, and an incidence at a vertex of B. Thus

$$t \left(\sum_{v \in [n] \setminus B} d_v - r^2 \Delta_1(\mathcal{G}) - 2\Delta_1(\mathcal{G})^2 \right) A_t \leq (r-1) d_y \left(\sum_{v \in B} d_v \right) A_{t-1}.$$

The Poisson comparison in Lemma 5.1 now applies. □

The next bound controls the cost of conditioning on the degree lower bounds.

Lemma 5.3. *For every fixed $r \geq 4$, if $(n, m, \boldsymbol{\ell})$ is near-critical, then a uniform m-edge r-graph $\mathcal{G}$ on $[n]$ satisfies*

$$\mathbb{P}[\mathcal{G} \in \Omega_r(n, m, \boldsymbol{\ell})] \geq \exp\{-n^{1/10}\} \tag{5.2}$$

for all sufficiently large n depending only on r.

Proof. First sample each r-set independently with probability $p_{\mathrm{low}} = (1 - n^{-1/4}) m / \binom{n}{r}$. Each vertex degree has mean $(1 + o(1)) \log n$, and Lemma 2.6(i) gives

$$\mathbb{P}_{p_{\mathrm{low}}}[d_{\mathcal{G}}(v) < \ell_v] \leq n^{-\varphi(0.01) + o(1)}.$$

Harris's inequality applies to the increasing events $\{d_{\mathcal{G}}(v) \geq \ell_v\}$. Their intersection consequently has probability at least

$$\prod_{v \in [n]} \mathbb{P}_{p_{\mathrm{low}}}[d_{\mathcal{G}}(v) \geq \ell_v] \geq \left(1 - n^{-\varphi(0.01) + o(1)}\right)^n \geq \exp\{-n^{1 - \varphi(0.01) + o(1)}\}.$$

The binomial number of edges exceeds m with probability at most $\exp\{-\Omega(n^{1/2} \log n)\}$ by Lemma 2.6(iii). The probability of satisfying all degree lower bounds in a uniform t-edge r-graph is nondecreasing in t. Hence

$$\mathbb{P}_{p_{\mathrm{low}}}[d_{\mathcal{G}}(v) \geq \ell_v \text{ for every } v] \leq \mathbb{P}[\mathcal{G} \in \Omega_r(n, m, \boldsymbol{\ell})] + \mathbb{P}_{p_{\mathrm{low}}}[|\mathcal{G}| > m].$$

Since $1 - \varphi(0.01) < 1/10$, subtracting the negligible second term proves (5.2). □

We now combine these switchings to obtain the degree bounds needed after deleting a bounded number of vertices.

Proposition 5.4. *Fix $r \geq 4$, and let $(n, m, \boldsymbol{\ell})$ be admissible and near-critical. Choose F_* uniformly in $\Omega_r(n, m, \boldsymbol{\ell})$, and put $B_* := \{v \in [n] \colon d_{F_*}(v) \leq 0.03 \log n\}$. There is a constant C_r such that, with probability $1 - o(1)$, the following statements hold.*

(i) We have $\Delta_1(F_*) \leq C_r \log n$, $|B_*| \leq n^{1/4}$, and $\Delta_2(F_*) \leq 2$.

(ii) For every $y \in [n]$, $\sum_{v \in B_* \setminus \{y\}} d_{F_*}(y, v) \leq 1$.

For every fixed integer $h \geq 0$, these two statements imply, for all sufficiently large n and every $Z \subseteq [n]$ with $|Z| \leq h$,

$$\sum_{v \in [n] \setminus Z} \max\{0, \ell_v - d_{F_* - Z}(v)\} \leq |Z|. \tag{5.3}$$

The convergence and the threshold on n depend only on r, h.

Proof. By Lemma 5.1, the degree above $\max\{\ell_v, \lceil rm/n \rceil\}$ is dominated by a Poisson variable of mean $rm/n \sim \log n$. Thus, for a sufficiently large C_r, the Poisson upper tail is at most n^{-3}, and a union bound gives $\mathbb{P}[\Delta_1(F_*) > C_r \log n] = o(1)$.

For the low-degree set, use independent edge sampling with probability $p = m/\binom{n}{r}$. Fix $S \subseteq [n]$ of size $t = \lceil n^{1/4} \rceil$. For each $v \in S$, count only the edges whose other vertices avoid S. These counts are independent and have mean $(1 + o(1)) \log n$. Therefore

$$\mathbb{P}_p[d_{\mathcal{G}}(v) \le 0.03 \log n \text{ for every } v \in S] \le n^{-(\varphi(0.03)+o(1))t}.$$

Since $\varphi(0.03) > 3/4$, summing over the $\binom{n}{t} \le (en/t)^t$ choices of S gives $\exp\{-\Omega(n^{1/4} \log n)\}$. Stirling's formula gives $\mathbb{P}_p[|\mathcal{G}| = m] = \Theta(m^{-1/2})$. Conditioning first on m edges and then on the degree lower bounds, using (5.2), still leaves an $o(1)$ failure probability.

Condition on any degree sequence satisfying the maximum-degree and low-degree-set bounds. In Lemma 5.2, take $B = \{x\}$ and $y \neq x$. Its Poisson mean is $O_r(\log n/n)$, uniformly in the conditioned sequence. Hence

$$\mathbb{P}[\Delta_2(F_*) > 2 \mid \boldsymbol{d}] \le \binom{n}{2} \frac{O_r((\log n/n)^3)}{3!} = o(1).$$

Taking $B = B_* \setminus \{y\}$ instead gives mean $O_r(n^{-3/4} \log n)$, so

$$\mathbb{P}\left[\exists y \in [n] \colon \sum_{v \in B_* \setminus \{y\}} d_{F_*}(y, v) > 1 \,\middle|\, \boldsymbol{d}\right] \le n \frac{O_r((n^{-3/4} \log n)^2)}{2!} = o(1).$$

These bounds are uniform over the conditioned degree sequences and prove (i)–(ii).

Finally, deleting at most h vertices removes at most $2h$ incident edges at any remaining vertex. A vertex outside B_* has degree greater than $0.03 \log n$, whereas its lower bound is at most $0.01 \log n + 10r$, so it still satisfies that bound for sufficiently large n. At the other vertices,

$$\sum_{v \in B_* \setminus Z} \max\{0, \ell_v - d_{F_* - Z}(v)\} \le \sum_{y \in Z} \sum_{v \in B_* \setminus \{y\}} d_{F_*}(y, v) \le |Z|.$$

This proves (5.3). □

6 Reduction to mixed Hamilton cycles

We first state the estimate for the random hypergraph left after the low-degree vertices have been treated. Fix $r \ge 4$. Let $(n_*, m_*, \boldsymbol{\ell})$ be admissible and near-critical as defined in Section 5. Before sampling the hypergraph, fix a matching M_0 of s prescribed pairs on $[n_*]$, where $1 \le s \le n_*^{1/10}$ and $(r-1) \mid n_* - s$. The set of *M_0-compatible edges* is

$$\mathcal{K}_{M_0} := \left\{ e \in \binom{[n_*]}{r} \colon |e \cap V(M_0)| \le 1 \right\}.$$

For this fixed matching, write $\Phi_{M_0}(H) := \Phi(H \cap \mathcal{K}_{M_0}, M_0)$. Every counted cycle contains exactly $(n_* - s)/(r-1)$ ordinary edges. In all auxiliary families, compatibility will continue to refer to $V(M_0)$, even when a prescribed pair is added or removed.

Let F_* be chosen uniformly from $\Omega_r(n_*, m_*, \boldsymbol{\ell})$.

Theorem 6.1. *Fix $r \ge 4$. There is a constant $C_r > 0$ such that, for every admissible near-critical $(n_*, m_*, \boldsymbol{\ell})$ and every matching M_0 chosen as above, with probability $1 - o(1)$,*

$$\log \Phi_{M_0}(F_*) \ge \frac{n_* - s}{r-1} \log \frac{r(r-1)m_*}{n_*} - (n_* - s) - C_r \frac{n_*}{\log n_*}.$$

The failure probability tends to zero uniformly over these choices.

The proof in Section 7 reduces Theorem 6.1 to the edge-marginal bound in Proposition 7.2, whose proof is given in Section 10. We first deduce Theorem 1.1 from Theorem 6.1. For the rest of this section, probabilities refer to the original process $(\mathcal{H}_t)$ on $[n]$.

The exposure below fixes the prescribed pairs before the core is sampled. We first identify the stopping-time bias that determines the conditional distribution of this core.

For an m-edge r-graph F with $\delta_1(F) \ge 1$, let $b(F)$ be the number of edges of F containing at least one vertex of degree exactly 1. Each such edge is counted once.

Lemma 6.2. *For every such F,*

$$\mathbb{P}[\mathcal{H}_{\tau_1} = F] = \frac{b(F)}{m\binom{\binom{n}{r}}{m}}. \tag{6.1}$$

Proof. Conditional on $\mathcal{H}_m = F$, the last added edge is uniform among the m edges of F. Since the minimum-degree condition is increasing, $\tau_1 = m$ exactly when deleting this last edge from F destroys the condition $\delta_1 \geq 1$. This occurs precisely when the edge contains a vertex of degree 1. Also $\mathbb{P}[\mathcal{H}_m = F] = \binom{\binom{n}{r}}{m}^{-1}$. Multiplying the two probabilities proves (6.1). □

Put $d_0 \coloneqq \lfloor 0.01 \log n \rfloor$. Let B be the vertices of degree at most d_0 at τ_1, and let T_B contain all edges of $\mathcal{H}_{\tau_1}$ meeting B. We reveal τ_1, B, and T_B, writing their values as

$$\tau_1 = m, \quad B \coloneqq \{v \colon d_{\mathcal{H}_{\tau_1}}(v) \leq d_0\}, \quad T_B \coloneqq \{e \in \mathcal{H}_{\tau_1} \colon e \cap B \neq \varnothing\}.$$

We first record the properties that allow disjoint replacements through these vertices.

Lemma 6.3. *With high probability,*

$$1 \leq |B| \leq n^{1/12}, \quad \Delta_1(\mathcal{H}_{\tau_1}) \leq C_r \log n, \quad \Delta_2(\mathcal{H}_{\tau_1}) \leq 2,$$

and no two distinct vertices of B are joined by a Berge path of length at most 4. Moreover,

$$\left| \frac{r\tau_1}{n} - \log n \right| \leq 2 \log\log n + o(1). \tag{6.2}$$

Proof. Assign independent uniform arrival times in $[0,1]$ to the r-sets, and let $\mathcal{H}(p)$ contain those arriving by time p. Take

$$p_{\text{low}} \coloneqq \frac{\log n - \log\log n}{\binom{n-1}{r-1}}, \quad p_{\text{high}} \coloneqq \frac{\log n + \log\log n}{\binom{n-1}{r-1}}.$$

The expected numbers of isolated vertices at these parameters are $(1+o(1))\log n$ and $(1+o(1))/\log n$, respectively. For two distinct vertices, the ratio of their joint isolation probability at p_{low} to the product of their isolation probabilities is

$$(1-p_{\text{low}})^{-\binom{n-2}{r-2}} = 1 + O_r(\log n/n).$$

The second moment method at p_{low} and the first moment method at p_{high} therefore give $\mathcal{H}(p_{\text{low}}) \subseteq \mathcal{H}_{\tau_1} \subseteq \mathcal{H}(p_{\text{high}})$ with high probability.

A vertex degree in $\mathcal{H}(p_{\text{low}})$ has mean $(1+o(1))\log n$. By Lemma 2.6(i),

$$\mathbb{P}[d_{\mathcal{H}(p_{\text{low}})}(v) \leq d_0] \leq n^{-\varphi(0.01)+o(1)}, \quad \varphi(0.01) > 0.9439.$$

On the sandwich event, B is contained in the set of vertices of degree at most d_0 in $\mathcal{H}(p_{\text{low}})$. The expected size of this latter set is at most $n^{1-\varphi(0.01)+o(1)} = o(n^{1/12})$, so Markov's inequality gives $|B| \leq n^{1/12}$ with high probability.

For each $1 \leq t \leq 4$, there are $O_r(n^{(r-1)t+1})$ labeled Berge paths of length t. This bound permits extra intersections between path edges. Fix such a path and condition on its edges being present in $\mathcal{H}(p_{\text{high}})$. At each endpoint, count only incident edges in $\mathcal{H}(p_{\text{low}})$ that are not path edges and avoid the other endpoint. The two families of edge indicators are disjoint and independent of the specified path edges, and each count has mean $(1+o(1))\log n$. If both endpoints have total degree at most d_0 in $\mathcal{H}(p_{\text{low}})$, these two counts are also at most d_0. Consequently the expected number of such paths is at most

$$O_r(n^{(r-1)t+1}) O_r\left(\frac{\log n}{n^{r-1}}\right)^t n^{-2\varphi(0.01)+o(1)} = o(1).$$

Together with the sandwich, this proves separation.

Binomial upper tails in $\mathcal{H}(p_{\text{high}})$ give the maximum degree bound. The expected number of pairs contained in at least 3 edges is at most

$$\binom{n}{2}\binom{\binom{n-2}{r-2}}{3} p_{\text{high}}^3 = O_r(\log^3 n/n) = o(1).$$

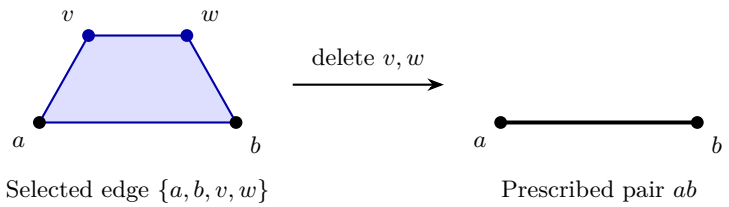


Figure 1: The low-degree reduction for $r = 4$. The selected edge through the exceptional vertex v supplies the prescribed pair ab after its entire private block $\{v, w\}$ is deleted.

The edge added at τ_1 contains a vertex whose degree becomes 1, so B is nonempty for large n. Finally, the edge counts at p_{low} and p_{high} differ from their means by $o(n)$ with high probability. Their means are $\binom{n}{r}p_{\text{low}}$ and $\binom{n}{r}p_{\text{high}}$, and $r\binom{n}{r}/n = \binom{n-1}{r-1}$. The sandwich therefore gives (6.2). □

Fix deterministic orders on the vertex labels and incident edges. For each $v \in B$, select its first incident edge e_v and the first two vertices a_v, b_v of $e_v \setminus \{v\}$. Let $P_v := e_v \setminus \{a_v, b_v\}$ be its private block, so $|P_v| = r - 2$ and $v \in P_v$. These choices are determined by (B, T_B). If the selected edges are not pairwise disjoint, the reduction stops; by Lemma 6.3, this happens with probability $o(1)$.

Let $s := |B|$ and let $D := \bigcup_{v \in B} P_v$ be the set of vertices to delete. Reveal every additional edge meeting D. Write $T_D := \{e \in \mathcal{H}_{\tau_1} : e \cap D \neq \varnothing\}$ for the revealed edge set and $W := [n] \setminus D$ for the remaining vertex set. Since $B \subseteq D$, this exposure includes T_B. The contraction is illustrated in Figure 1. The remaining hypergraph has $n_* = n - (r-2)s$ vertices and $m_* = m - |T_D|$ edges. For every $w \in W$, the revealed edges contribute $d_{T_D}(w)$ to its degree, leaving the lower bound $\ell_w := \max\{0, d_0 + 1 - d_{T_D}(w)\}$. Identify W with $[n_*]$ in increasing order of its labels.

Proposition 6.4. *Conditional on any $\mathbb{P}$-admissible exposure above for which the selected edges are pairwise disjoint, $\mathcal{H}_{\tau_1}[W]$ is uniform in $\Omega_r(n_*, m_*, \boldsymbol{\ell})$. With high probability, the exposed parameters satisfy (5.1) with offset allowance $10r$, and $M_0 := \{a_v b_v : v \in B\}$ is a matching satisfying $1 \le s \le n_*^{1/10}$ and $(r-1) \mid n_* - s$.*

Proof. For large n, every vertex of degree 1 belongs to B, and all its incident edges are recorded in T_B. Thus $b(F)$ in (6.1) is constant over all completions of the exposure. The set D and the selected subgraphs are determined by (B, T_B), and every degree of a deleted vertex is completely recorded in T_D. Every remaining vertex $w \in W$ lies outside B, which is equivalent to

$$d_{\mathcal{H}_{\tau_1}[W]}(w) + d_{T_D}(w) \ge d_0 + 1.$$

Conversely, every m_*-edge r-graph on W satisfying these inequalities completes the same exposure. The degrees of vertices in D are already fixed, and all unexposed edges lie inside W. The feasible completions are therefore exactly $\Omega_r(n_*, m_*, \boldsymbol{\ell})$. Since their total edge count is fixed, (6.1) assigns them equal probabilities.

A remaining vertex cannot belong to edges meeting private blocks of two different vertices of B. Otherwise those edges, together with the two selected edges, give a Berge path of length at most 4 between the exceptional vertices. Within a single block, the codegree bound gives $d_{T_D}(w) \le 2(r-2)$. Also $n_* = n - O_r(n^{1/12})$, so $|\lfloor 0.01 \log n \rfloor - \lfloor 0.01 \log n_* \rfloor| \le 1$ for large n. Hence $\ell_w = \lfloor 0.01 \log n_* \rfloor + O_r(1)$, with error at most $10r$.

The maximum degree bound gives $|T_D| = O_r(s \log n)$. Since $m_* = \tau_1 - |T_D|$ and $n_* = n - (r-2)s$,

$$\frac{r m_*}{n_*} - \frac{r\tau_1}{n} = O_r(s \log n / n) = o(1).$$

Thus (6.2) implies the density window in (5.1). The pairs form a matching because the selected subgraphs are disjoint. Also $n_* - s = n - (r-1)s$, which is divisible by $r-1$. Finally, $s \le n^{1/12}$ and $n_* = n - O_r(n^{1/12})$ give $s \le n_*^{1/10}$ for large n. □

Proof of Theorem 1.1 assuming Theorem 6.1. Replace each prescribed pair $a_v b_v$ by its selected edge e_v. These expansions preserve a single cyclic component and cover every deleted vertex exactly once. They

are injective because the selected edges are fixed by the exposure, and contracting them recovers the original mixed cycle. Apply Theorem 6.1 conditional on each exposure whose parameters have the stated bounds, and average its uniform failure probability over these exposures. Since their bounds hold with high probability, this proves Theorem 1.1. The conditioning in this application is on the exposed data alone. Separation is used to establish the parameter bounds, rather than imposed as an additional condition on the unexposed core. □

7 Counting mixed Hamilton cycles by edge deletion

We prove Theorem 6.1 assuming the edge-marginal proposition stated below. The edge-marginal bound controls the loss from each deletion. A martingale estimate controls the sum of these losses up to the first state where the marginal bound is unavailable.

For F_* from Section 6, let $B_* := \{v \in [n_*] \colon d_{F_*}(v) \le 0.03 \log n_*\}$. We use the following laws on edge-deletion sequences.

- $\mathbb{P}$ is the law of the sequence

$$\binom{[n_*]}{r} = H_{\binom{n_*}{r}} \supsetneq \cdots \supsetneq H_{m_*+1} \supsetneq H_{m_*},$$

  where each step deletes a uniformly chosen edge.
- $\mathbb{P}^*$ is the same law conditional on $H_{m_*} \in \Omega_r(n_*, m_*, \boldsymbol{\ell})$, so $H_{m_*} = F_*$. Equivalently, sample F_* uniformly from this family and add the missing edges in a uniformly random order.

Write β for the probability of this conditioning event; that is,

$$\beta := \mathbb{P}[H_{m_*} \in \Omega_r(n_*, m_*, \boldsymbol{\ell})] = \frac{|\Omega_r(n_*, m_*, \boldsymbol{\ell})|}{\binom{\binom{n_*}{r}}{m_*}} \ge \exp\{-n_*^{1/10}\},$$

where the inequality follows from Lemma 5.3.

Throughout the edge-deletion argument, let $k = (n_* - s)/(r-1)$ denote the number of ordinary edges in a M_0-respecting mixed Hamilton cycle. The reference value for the logarithm of $\Phi_{M_0}(H_j)$ is

$$L_j := \log \Phi_{M_0}\left(\binom{[n_*]}{r}\right) - k \sum_{h=j+1}^{\binom{n_*}{r}} \frac{1}{h}. \tag{7.1}$$

The following calculation identifies its main term.

Lemma 7.1. *For every $m_* \le j \le \binom{n_*}{r}$,*

$$L_j = k \log \frac{r(r-1)j}{n_*} - (r-1)k + O_r(s \log n_* + \log n_*). \tag{7.2}$$

Proof. By Lemma 2.3, requiring every ordinary edge to be M_0-compatible changes the complete-hypergraph cycle number by a factor $1 - o(1)$. Stirling's formula, together with $n_* = (r-1)k + s$ and $(r-1)! = (r-1)(r-2)!$, gives

$$\log \Phi_{M_0}\left(\binom{[n_*]}{r}\right) = k \log \left((r-1)\binom{n_*-1}{r-1}\right) - (r-1)k + O_r(s \log n_* + \log n_*).$$

The harmonic sum in (7.1) is $\log\big(\binom{n_*}{r}/j\big) + O(1/j)$. Since $k/j = O_r(1/\log n_*)$, subtracting this sum and using $\binom{n_*-1}{r-1} = r\binom{n_*}{r}/n_*$ proves (7.2). □

We next define the edge marginal, which measures the fraction of counted mixed Hamilton cycles containing a specified edge. When $\Phi_{M_0}(H) > 0$, put

$$q_e(H) := \frac{|\{C \colon C \text{ is counted by } \Phi_{M_0}(H),\ e \in E_r(C)\}|}{\Phi_{M_0}(H)}, \quad \sum_{e \in H} q_e(H) = k,$$

with $q_e = 0$ for forbidden edges, and define

$$\mathsf{A}_j := \{\log \Phi_{M_0}(H_j) \geq L_j - n_*/\sqrt{\log n_*}\}.$$

The following proposition bounds every edge marginal whenever the counting lower bound holds. Its proof is given in Section 10.

Proposition 7.2. *There exists a constant $C_r > 0$ such that, with $\mathbb{P}^*$-probability $1 - o(1)$, simultaneously for every $m_* \leq j \leq \binom{n_*}{r}$ for which A_j holds, we have*

$$\max_{e \in H_j} q_e(H_j) \leq \frac{C_r k}{j}.$$

Proof of Theorem 6.1 assuming Proposition 7.2. Choose $C_0 \geq 1$ at least the constant in Proposition 7.2. Define the variance bound

$$\sigma^2 := C_0^2 k^2 \sum_{j=m_*+1}^{\binom{n_*}{r}} \frac{1}{j^2} = O_r\left(\frac{n_*}{\log n_*}\right),$$

where the estimate follows from $\sum_{j>m_*} j^{-2} \leq 1/m_*$, $k \asymp_r n_*$, and $m_* \asymp_r n_* \log n_*$.

Starting from the complete hypergraph, stop at the first visited state H_j for which $\Phi_{M_0}(H_j) = 0$ or $\max_{e \in H_j} q_e(H_j) > C_0 k/j$. Let j_0 be the size of this state, or m_* if no earlier stop occurs. Thus every deletion from a size $h > j_0$ is made from a state satisfying the marginal bound, while the estimate below also includes the state H_{j_0} reached by the last such deletion.

Claim 7.3. *For every $a > 0$,*

$$\mathbb{P}^*\left[\min_{j_0 \leq j \leq \binom{n_*}{r}} \{\log \Phi_{M_0}(H_j) - L_j\} < -a - \sigma^2\right] \leq \beta^{-1} \exp\left\{-\frac{a^2}{2\sigma^2}\right\}.$$

Proof. Work first under $\mathbb{P}$. Let $\mathcal{F}_j := \sigma(H_{\binom{n_*}{r}}, \dots, H_j)$ record the deletions down to size j. As j decreases, these sigma-algebras form a filtration: $\mathcal{F}_j \subseteq \mathcal{F}_{j-1}$. Whether deletion continues from H_j is determined by $\mathcal{F}_j$. If it continues, let e_j be the uniformly deleted edge and put $D_j := q_{e_j}(H_j) - k/j$; otherwise put $D_j = 0$ and set every later increment to zero. Since deletion is uniform under $\mathbb{P}$,

$$\mathbb{E}[q_{e_j}(H_j) \mid \mathcal{F}_j] = \frac{1}{j} \sum_{e \in H_j} q_e(H_j) = \frac{k}{j}$$

whenever deletion continues. Consequently $\mathbb{E}[D_j \mid \mathcal{F}_j] = 0$ and $|D_j| \leq C_0 k/j$. Hoeffding's exponential-moment bound [17] gives

$$\mathbb{E}[\exp\{\lambda D_j\} \mid \mathcal{F}_j] \leq \exp\left\{\frac{\lambda^2 C_0^2 k^2}{2j^2}\right\} \quad \text{for every } \lambda > 0.$$

Let S_t be the sum of the first t increments, in decreasing order of the sizes, and let v_t be the corresponding sum of $C_0^2 k^2/j^2$. The process $\exp\{\lambda S_t - \lambda^2 v_t/2\}$ is a nonnegative supermartingale with initial value one. Its maximal inequality, the usual proof of the maximal Azuma–Hoeffding bound [2], yields

$$\mathbb{P}[\max_t S_t > a] \leq \exp\{-\lambda a + \lambda^2 \sigma^2/2\}.$$

Taking $\lambda = a/\sigma^2$ gives the upper bound $\exp\{-a^2/(2\sigma^2)\}$.

For $h > j_0$, the exact counting identity is $\Phi_{M_0}(H_{h-1}) = \Phi_{M_0}(H_h)(1 - q_{e_h}(H_h))$. Since $C_0 k/m_* \leq 1/2$ for large n_*, the marginal bound ensures that the count stays positive. Also $-\log(1-q) \leq q + q^2$ for $0 \leq q \leq 1/2$. Therefore, for every $j \geq j_0$,

$$\begin{aligned}\log \Phi_{M_0}(H_j) - L_j &= \sum_{h=j+1}^{\binom{n_*}{r}} \left(\log(1 - q_{e_h}(H_h)) + \frac{k}{h}\right) \\ &\geq -\sum_{h=j+1}^{\binom{n_*}{r}} D_h - C_0^2 k^2 \sum_{h=j+1}^{\binom{n_*}{r}} \frac{1}{h^2}.\end{aligned}$$

Outside the maximal martingale event, this is at least $-a-\sigma^2$. Finally, for every event E determined by the deletion sequence,

$$\mathbb{P}^*[E] = \frac{\mathbb{P}[E \cap \{H_{m_*} \in \Omega_r(n_*, m_*, \boldsymbol{\ell})\}]}{\beta} \le \beta^{-1}\mathbb{P}[E].$$

This transfers the estimate to $\mathbb{P}^*$ and proves the claim. □

Take $a = n_*/\log n_*$. By (5.2), the failure probability in Claim 7.3 is at most

$$\beta^{-1}\exp\left\{-\frac{a^2}{2\sigma^2}\right\} \le \exp\{-c_r n_*/\log n_*\} = o(1).$$

Intersect this success event with the event in Proposition 7.2, on which the marginal bound holds whenever A_j holds. This intersection has $\mathbb{P}^*$-probability $1-o(1)$. Suppose A_j first fails at size j. Every preceding state satisfies the counting lower bound and hence the marginal bound. Thus $j \ge j_0$, and Claim 7.3 gives

$$\log \Phi_{M_0}(H_j) \ge L_j - O_r(n_*/\log n_*),$$

contradicting the failure threshold $n_*/\sqrt{\log n_*}$. Since $\mathsf{A}_{\binom{n_*}{r}}$ holds at the complete hypergraph, A_j holds throughout the sequence down to m_*. The same estimate at m_*, together with Lemma 7.1 and $s\log n_* = o(n_*/\log n_*)$, proves Theorem 6.1. □

8 Conditional laws under edge deletion

We now return to the edge-deletion laws $\mathbb{P}, \mathbb{P}^*$ defined in Section 7 and describe the conditional distributions used below. Under $\mathbb{P}^*$, for every fixed $j \in [m_*, \binom{n_*}{r}]$, the pair (F_*, H_j) is uniform among

$$F_* \subseteq H_j \subseteq \binom{[n_*]}{r}, \quad |F_*| = m_*, \quad |H_j| = j, \quad F_* \in \Omega_r(n_*, m_*, \boldsymbol{\ell}).$$

Put $Z_{m_*}(H) := |\{F \subseteq H : F \in \Omega_r(n_*, m_*, \boldsymbol{\ell})\}|$. Under $\mathbb{P}^*$, the probability of $H_j = H$ is proportional to $Z_{m_*}(H)$. Consequently, the next deleted edge need not be uniform in H. For $j > m_*$, $|H| = j$, $Z_{m_*}(H) > 0$, and $e \in H$,

$$\mathbb{P}^*[e \text{ is deleted next} \mid H_j = H] = \frac{Z_{m_*}(H-e)}{(j-m_*)Z_{m_*}(H)}.$$

Indeed, conditional on $H_j = H$, the hypergraph F_* is uniform among the $Z_{m_*}(H)$ feasible subgraphs of H, and the next deleted edge is uniform in its complement. Each feasible subgraph leaves $j-m_*$ edges available for deletion. The following lemmas give the corresponding laws after fixed exposures and auxiliary edge deletions.

Fix a set $D \subseteq [n_*]$ of bounded size and a time $m_* \le j \le \binom{n_*}{r}$. The edges meeting D are

$$F_*(D) := \{e \in F_* : e \cap D \neq \varnothing\}, \quad H_j(D) := \{e \in H_j : e \cap D \neq \varnothing\}.$$

Write $\hat{F}_* := F_* - D$ and $\hat{H}_j := H_j - D$. For every $v \in [n_*] \setminus D$, the remaining degree lower bound is $\ell'_v := \max\{0, \ell_v - d_{F_*(D)}(v)\}$. We reveal the two incident edge sets $F_*(D), H_j(D)$, leaving $\hat{F}_*, \hat{H}_j$ unrevealed.

Lemma 8.1. *Conditional on any $\mathbb{P}^*$-admissible values of $F_*(D), H_j(D)$, the following statements hold.*

(i) $\hat{F}_*$ is uniform among the r-graphs on $[n_*] \setminus D$ with $m_* - |F_*(D)|$ edges and degrees at least ℓ'_v at every vertex v.

(ii) Conditional on $\hat{F}_*$, the hypergraph $\hat{H}_j$ is uniform among its extensions on $[n_*] \setminus D$ with $j - |H_j(D)|$ edges.

(iii) For every $y \in [n_*] \setminus D$,

$$\sum_{v \in [n_*] \setminus (D \cup \{y\})} \max\{0, \ell'_v - d_{\hat{F}_* - y}(v)\} \le \sum_{v \in [n_*] \setminus \{y\}} \max\{0, \ell_v - d_{F_* - y}(v)\}.$$

If F_* satisfies Proposition 5.4(i)–(ii), then the left side is at most one.

Proof. Restriction gives a bijection between original nested pairs with the specified incident edge sets and nested pairs on $[n_*] \setminus D$ satisfying the stated edge counts and degree lower bounds. Its inverse adds $F_*(D)$ and $H_j(D)$ to the inner and outer hypergraphs, respectively. The constraints at vertices of D are already determined by $F_*(D)$. All original pairs have equal probability, and each possible inner hypergraph has exactly

$$\binom{\binom{n_*-|D|}{r} - (m_* - |F_*(D)|)}{j - |H_j(D)| - m_* + |F_*(D)|}$$

outer extensions. This number depends only on the fixed edge counts, proving (i)–(ii).

For (iii), let $c_D(v, y)$ count the edges of F_* containing v, y and avoiding D. If $\ell_v \le d_{F_*(D)}(v)$, then $\ell'_v = 0$, so deleting y cannot violate the remaining lower bound at v. Otherwise,

$$\max\{0, \ell'_v - d_{\hat{F}_* - y}(v)\} = \max\{0, \ell_v - d_{F_*}(v) + c_D(v, y)\} \le \max\{0, \ell_v - d_{F_*}(v) + d_{F_*}(v, y)\}.$$

Summing gives the comparison, and (5.3) with $Z = \{y\}$ gives the final assertion. □

Lemma 8.2. *Let $\mathcal{E}$ be a finite edge universe, let $\mathcal{U} \subseteq \mathcal{E}$, and let $\mathfrak{F} \subseteq 2^{\mathcal{E}}$ be an increasing family. Suppose that (S_*, H) is uniform among*

$$S_* \subseteq H \subseteq \mathcal{E}, \quad |S_*| = m_*, \quad |H| = j, \quad S_* \in \mathfrak{F}.$$

Conditional on any positive-probability record $A_0 := S_* \setminus \mathcal{U}$ and $B_0 := H \setminus \mathcal{U}$, the pair $(S, J) := (S_* \cap \mathcal{U}, H \cap \mathcal{U})$ is uniform among

$$S \subseteq J \subseteq \mathcal{U}, \quad |S| = m_* - |A_0|, \quad |J| = j - |B_0|, \quad A_0 \cup S \in \mathfrak{F}.$$

Proof. After the outside record is fixed, the restriction map is a bijection with inverse $(S, J) \mapsto (A_0 \cup S, B_0 \cup J)$. It therefore preserves equal probabilities. □

The next lemma identifies the distribution of the deleted set T given the two remaining hypergraphs. Feasibility of the remaining inner hypergraph makes every reverse extension admissible.

Lemma 8.3. *In the conditional experiment of Lemma 8.2, fix $0 \le t \le |J|$ and choose T uniformly from the t-subsets of J. Put $F := J \setminus T$, $F_0 := S \setminus T$, and $T_0 := S \cap T$. Conditional on possible values of F, F_0, A_0, B_0, the pair (T_0, T) is uniform among*

$$T_0 \subseteq T \subseteq \mathcal{U} \setminus F, \quad |T| = t, \quad |T_0| = m_* - |A_0| - |F_0|, \quad A_0 \cup F_0 \cup T_0 \in \mathfrak{F}.$$

If $A_0 \cup F_0 \in \mathfrak{F}$, then

$$T \mid F, F_0, A_0, B_0 \ \sim \ \mathrm{Unif}\binom{\mathcal{U} \setminus F}{t}. \tag{8.1}$$

Proof. The sizes of S, J, T are fixed, so every admissible forward triple (S, J, T) has the same probability. Given F, F_0, its inverse description is $S = F_0 \cup T_0$ and $J = F \cup T$. These formulas give a bijection between forward triples with the specified remainders and the pairs displayed in the statement.

Put $a := m_* - |A_0| - |F_0|$. If $A_0 \cup F_0 \in \mathfrak{F}$, monotonicity permits every a-subset T_0 of every t-subset T of $\mathcal{U} \setminus F$. Hence each possible T has exactly $\binom{t}{a}$ preimages. Since this number is independent of T, the marginal law of T is (8.1). □

To use the uniform reverse law in Lemma 8.3, we need the smaller remaining hypergraph to satisfy its degree lower bounds. The next lemma shows that a small uniform set of deleted edges preserves these bounds with high probability. Here the pair $F_* \subseteq H$ is fixed, and probability concerns only the choice of T.

Lemma 8.4. *Fix a pair $F_* \subseteq H$ satisfying Proposition 5.4(i)–(ii). Let $\mathcal{U} \subseteq H$ have size $m_0 \ge 1$, let $k \succ_r n_*$, and let $1 \le \nu \le \log n_*$. Choose T uniformly from the subsets of $\mathcal{U}$ of size $\lfloor \nu m_0/k \rfloor$. Then*

$$\mathbb{P}[F_* - T \text{ violates a degree lower bound } \ell_v] \le O_r(\nu n_*^{-3/4} \log n_*) + \exp\{-\Omega_r((\log n_*)^2)\}. \tag{8.2}$$

After a bounded vertex set D has been deleted and its incident edges of F_* exposed, the same bound holds for violation of the adjusted lower bounds in Lemma 8.1, provided every edge of $\mathcal{U}$ avoids D.

Proof. At most $C_r n_*^{1/4} \log n_*$ edges of F_* meet B_*. Each eligible edge enters T with probability $|T|/m_0 \le \nu/k = O_r(\nu/n_*)$. The probability that the set T contains one of these edges is therefore $O_r(\nu n_*^{-3/4} \log n_*)$. Otherwise every vertex in B_* retains its degree in F_*.

For a vertex outside B_*, the degree in F_* exceeds $0.03 \log n_*$, while $\ell_v = 0.01 \log n_* + O_r(1)$. For sufficiently large n_*, failure requires deletion of at least $a_{n_*} := \lfloor 0.01 \log n_* \rfloor$ incident edges of F_*. Any specified a_{n_*} eligible edges all enter T with probability at most $(|T|)_{a_{n_*}}/(m_0)_{a_{n_*}} \le (|T|/m_0)^{a_{n_*}}$. Since $\Delta_1(F_*) \le C_r \log n_*$, the failure probability at one vertex is at most

$$\binom{C_r \log n_*}{a_{n_*}} \left(\frac{C_r \nu}{n_*}\right)^{a_{n_*}} = \exp\{-\Omega_r((\log n_*)^2)\}.$$

If $|T| < a_{n_*}$, this probability is zero. Summing over vertices proves (8.2).

When every edge of T avoids D, it leaves all exposed edges meeting D unchanged. At a vertex outside D, subtracting the degree supplied by those edges converts the original lower bound exactly into its adjusted lower bound. Thus the same estimate applies. □

We next study the links $L_{\hat{F}_*}(y) \subseteq L_{\hat{H}_j}(y)$ at a vertex $y \in [n_*] \setminus D$. We reveal $F_*(D), H_j(D), \hat{F}_* - y, \hat{H}_j - y$, leaving these two links unrevealed. The edge counts of the revealed hypergraphs determine the sizes of the two links.

After deleting y, a vertex may be one edge below its adjusted degree lower bound. In that case, the smaller link must contain an edge through that vertex. The next lemma compares a uniform nested pair of sets with the pair conditioned to meet a fixed set; we will apply it to the two links at y.

Lemma 8.5. *Let $0 \le b \le q \le |E|$, let I_0 be a uniform b-subset of a finite set E, and let J_0 be its uniform extension to a q-subset. Fix $S \subseteq E$ with $\mathbb{P}[I_0 \cap S \ne \varnothing] > 0$. There is a coupling with a pair (I_1, J_1) having the law of (I_0, J_0) conditional on $I_0 \cap S \ne \varnothing$, such that I_1 and J_1 each differ from their unconditioned counterparts by at most one replacement.*

Proof. Put $Z := |I_0 \cap S|$. Its hypergeometric masses are log-concave, since the consecutive ratios

$$\frac{\mathbb{P}[Z = h+1]}{\mathbb{P}[Z = h]} = \frac{(|S| - h)(b - h)}{(h+1)(|E| - |S| - b + h + 1)}$$

decrease on their support. For $s_h := \mathbb{P}[Z \ge h]$, log-concavity implies $s_h/s_{h-1} \le s_1$ when $\mathbb{P}[Z = 0] > 0$. Indeed, $s_h/\mathbb{P}[Z = h]$ is a sum of products of consecutive mass ratios and is nonincreasing in h, so the hazard rate is nondecreasing. It follows that

$$Z \preceq (Z \mid Z \ge 1) \preceq Z + 1.$$

If $Z \ge 1$ almost surely, the conditioning has no effect. Otherwise a common-quantile coupling gives $Z_0 \le Z_1 \le Z_0 + 1$.

Choose independent uniform orders of S and $E \setminus S$. For $i \in \{0, 1\}$, take the first Z_i elements of S and the first $b - Z_i$ of its complement to form I_i. These sets have the required marginal laws and differ by at most one replacement. Let π be the transposition taking I_0 to I_1, or the identity when they agree. Conditional on I_0, sample its uniform extension J_0 using fresh randomness and set $J_1 := \pi(J_0)$. For each coupled pair I_0, I_1, the permutation π is a bijection between their extension spaces. Thus J_1 is a uniform extension of I_1 and differs from J_0 by at most one replacement. □

The next lemma applies Lemma 8.5 to these links. It shows that a small excluded family of $(r-1)$-sets, determined by the information outside y, is unlikely to contain most of $L_{\hat{H}_j}(y)$.

Lemma 8.6. *Fix $c > 0$, a bounded deleted set D, a time $m_* \le j \le \binom{n_*}{r}$, and a vertex $y \in [n_*] \setminus D$. Condition on $\mathbb{P}^*$-admissible values of $F_*(D), H_j(D), \hat{F}_* - y, \hat{H}_j - y$. Suppose that*

$$\sum_{v \in [n_*] \setminus (D \cup \{y\})} \max\{0, \ell'_v - d_{\hat{F}_* - y}(v)\} \le 1, \quad d_{\hat{H}_j}(y) \ge c \log n_*.$$

Let $\Gamma \subseteq \binom{[n_*]\setminus(D\cup\{y\})}{r-1}$ be determined by these revealed hypergraphs and fixed labels. If its density is at most $\rho = o(1)$, then

$$\mathbb{P}^*\left[|L_{\hat{H}_j}(y)\cap\Gamma| \ge \frac{2}{3}d_{\hat{H}_j}(y)\,\middle|\, F_*(D), H_j(D), \hat{F}_* - y, \hat{H}_j - y\right] \le (C\rho)^{d_{\hat{H}_j}(y)/4}. \tag{8.3}$$

Here C and the threshold on n_* depend only on r, c and the fixed bound on $|D|$.

Proof. Put $b := d_{\hat{F}_*}(y)$ and $q := d_{\hat{H}_j}(y)$. By Lemma 8.1, the links $L_{\hat{F}_*}(y)$ and $L_{\hat{H}_j}(y)$ have the uniform nested law with sizes b, q, subject to the adjusted lower bounds. The constraint at y is determined by b and is satisfied because the specified data have positive probability. At the other vertices there are two cases.

(i) If the total adjusted deficit is zero, the two links at y are an unconditioned uniform nested pair.

(ii) If the total adjusted deficit is one, a unique vertex w requires an edge of $\hat{F}_*$ containing both y and w. The two links at y are therefore a uniform nested pair conditioned on the smaller link meeting the fixed star $\{A \in \binom{[n_*]\setminus(D\cup\{y\})}{r-1} : w \in A\}$.

In both cases, Lemma 8.5 couples the link $L_{\hat{H}_j}(y)$ to a uniform q-subset of its universe with at most one replacement. For large n_*, the event in (8.3) consequently forces this uniform subset to contain at least $q/2$ elements of Γ. Its hypergeometric upper tail is at most $(2e\rho)^{q/2} \le (C\rho)^{q/4}$. If $\rho = 0$, the event is empty. □

We next control the degrees and edge distribution throughout the conditioned edge-deletion process.

Proposition 8.7. *Fix integers $h \ge 0$ and a constant $L > 0$. There are positive constants $c_{r,h,L}, C_{r,h,L}$ such that, with $\mathbb{P}^*$-probability $1 - o(1)$, the following statements hold simultaneously for every $m_* \le j \le \binom{n_*}{r}$.*

(i) For every $v \in [n_*]$,

$$c_{r,h,L}\frac{rj}{n_*} \le d_{H_j}(v) \le C_{r,h,L}\frac{rj}{n_*}.$$

(ii) The codegrees satisfy

$$\Delta_2(H_j) \le C_{r,h,L}\frac{rj}{n_*(\log n_*)^{1/4}}.$$

(iii) For every $A \subseteq [n_*]$ with $\big||A| - n_*/(r-1)\big| \le Ln_*^{1/10}$,

$$\big||H_j[A,\overline{A}]| - \theta_r j\big| \le C_{r,h,L}\frac{rj}{(\log n_*)^{1/8}}. \tag{8.4}$$

The same conclusions hold for every $H_j - D$ with $|D| \le h$, replacing n_*, j by $n_* - |D|, |H_j - D|$, respectively. Requiring the edges to be M_0-compatible also preserves the upper bounds and the edge-distribution estimate.

The last assertion concerns upper bounds and edge distribution; the compatible subhypergraph need not satisfy the vertex-degree lower bound.

Proof. Condition first on an F_* satisfying Proposition 5.4(i). Given F_*, the edges of $H_j \setminus F_*$ form a uniform $(j - m_*)$-subset of the missing r-sets. Thus the new incidences at any vertex are hypergeometric, and

$$\mathbb{E}[d_{H_j}(v) - d_{F_*}(v) \mid F_*] = \frac{(j-m_*)(\binom{n_*-1}{r-1} - d_{F_*}(v))}{\binom{n_*}{r} - m_*} \le \frac{2rj}{n_*}$$

for sufficiently large n_*. Since $d_{F_*}(v) \le C_r \log n_* = O_r(j/n_*)$, Lemma 2.6(ii), with a sufficiently large degree-bound constant, makes the upper-tail probability at most n_*^{-r-3}. Summing over the at most $n_*\binom{n_*}{r}$ vertex–time pairs proves the upper bound in (i).

For $j \le L_0 m_*$, where L_0 is a sufficiently large fixed constant, the lower bound follows from $d_{F_*}(v) \ge \ell_v = (0.01 + o(1))\log n_*$. For $j > L_0 m_*$, the preceding expectation is at least $rj/(2n_*)$. Applying Lemma 2.6(i) to the new incidences gives

$$\mathbb{P}^*\left[d_{H_j}(v) - d_{F_*}(v) < \frac{rj}{4n_*}\,\middle|\, F_*\right] \le \exp\{-c_r j/n_*\} \le n_*^{-r-3},$$

after increasing L_0. The same union bound proves the lower bound.

The new incidences of a pair have conditional mean at most $C_r rj/n_*^2$, and their original contribution is at most two. With $t = C_{r,h,L} rj/(n_*(\log n_*)^{1/4})$, the upper-tail bound therefore gives

$$\mathbb{P}^*[\Delta_2(H_j) > t \text{ for some } j \mid F_*] \le \binom{n_*}{2}\binom{n_*}{r} \exp\{-c_r(\log n_*)^{7/4}\} = o(1).$$

This proves (ii).

For (iii), work under the unconditioned deletion law $\mathbb{P}$ and first prove the estimate for the wider range $\big||A| - n_*/(r-1)\big| \le (L+1)n_*^{1/10}$. For a fixed A, the variable $|H_j[A, \overline{A}]|$ is hypergeometric with mean

$$j\frac{\binom{|A|}{2}\binom{n_*-|A|}{r-2}}{\binom{n_*}{r}} = \theta_r j + O_{r,L}(jn_*^{-9/10}).$$

The error is smaller than the tolerance in (8.4). The two-sided Chernoff bound consequently yields

$$\mathbb{P}\left[\big||H_j[A,\overline{A}]| - \theta_r j\big| > C_{r,h,L}\frac{rj}{(\log n_*)^{1/8}}\right] \le 2\exp\left\{-\frac{c_{r,L}j}{(\log n_*)^{1/4}}\right\}.$$

A union bound over A, j, followed by conditioning on $H_{m_*} \in \Omega_r(n_*, m_*, \boldsymbol{\ell})$, gives

$$\mathbb{P}^*[\text{(iii) fails}] \le 2\beta^{-1}2^{n_*}\binom{n_*}{r}\exp\left\{-\frac{c_{r,L}m_*}{(\log n_*)^{1/4}}\right\} = o(1),$$

since $m_* \asymp_r n_* \log n_*$ and $\beta^{-1} \le \exp\{n_*^{1/10}\}$.

Deleting at most h vertices changes a remaining vertex degree by at most $h\Delta_2(H_j) = o_r(j/n_*)$ and removes at most $h\Delta_1(H_j) = O_{r,h,L}(j/n_*)$ edges. The wider range for A includes every $A \subseteq [n_*] \setminus D$ with $\big||A| - (n_* - |D|)/(r-1)\big| \le L(n_* - |D|)^{1/10}$, because deleting D changes the size condition by $O_{r,h}(1)$. The deletion errors can be absorbed into (i)–(iii), including the changed centering and vertex count. Finally, at most $2s\Delta_1(H_j) = O_{r,h,L}(sj/n_*)$ edges fail to be M_0-compatible. Since $s \le n_*^{1/10}$, their removal preserves (iii), and the degree and codegree upper bounds can only decrease. □

9 Edge resampling for mixed Hamilton cycles

We now use an auxiliary set of edges to compare completion counts for present and absent edges. The forward argument preserves most abnormal completions, while the reverse argument samples uniformly from the missing edges. We first bound the variance of the number of surviving members of an indexed family.

Lemma 9.1. *Let $\mathcal{A}$ be a nonempty indexed family of h-subsets of an m-element universe, where $1 \le h \le m/4$. Let T be a uniform t-subset, with $0 \le t \le m/4$, and let Z count indexed members avoiding T. Put $\zeta_h := \binom{m-h}{t}/\binom{m}{t}$, and let A_1, A_2 be independent uniform indexed members. Then*

$$\mathbb{E}[Z] = \zeta_h|\mathcal{A}|, \quad \frac{\mathrm{Var}[Z]}{(\mathbb{E}[Z])^2} \le \left(\exp\left\{\frac{4th}{m}\right\} - 1\right)\frac{\mathbb{E}[|A_1 \cap A_2|]}{h}.$$

Proof. For a fixed pair, put $I := |A_1 \cap A_2|$. It survives jointly with probability $\binom{m-2h+I}{t}/\binom{m}{t}$. Dividing by ζ_h^2 gives

$$\prod_{i=0}^{t-1}\frac{(m-2h+I-i)(m-i)}{(m-h-i)^2}.$$

The excess of the ith factor over 1 is $[I(m-i) - h^2]/(m-h-i)^2 \le 4I/m$, so the product is at most $\exp\{4tI/m\}$. Convexity on $0 \le I \le h$ gives

$$\exp\left\{\frac{4tI}{m}\right\} \le 1 + \left(\exp\left\{\frac{4th}{m}\right\} - 1\right)\frac{I}{h}.$$

Average over the ordered pair of indexed members and subtract 1. This calculation retains their multiplicities. □

Fix the constants in Proposition 8.7 with $h = 4r$ and $L = 2$. References to the process bounds below mean all conclusions of that proposition with these choices.

From this section onward, $\Psi_{H,M}(P; u \to v)$ counts only cycles whose ordinary edges belong to $H \cap \mathcal{K}_{M_0}$. The average degree $d(H)$ continues to use all edges of H.

Definition 9.2. *A* frame Λ *consists of the following data.*

(i) A deleted set $\hat{B} \subseteq [n_*]$ with $|\hat{B}| \le 4r$. Write $V_\Lambda := [n_*] \setminus \hat{B}$.

(ii) A nonempty prescribed matching M on V_Λ, obtained from M_0 by removing at most two pairs and adding at most two pairs. We require $(r-1) \mid |V_\Lambda| - |M|$.

(iii) A root pair of M with a fixed direction.

We consider the deterministic catalogue of all such labelled frames. There are $n_*^{O_r(1)}$ frames: the deleted vertices, removed pairs, added pairs, and directed root have at most n_*^{4r}, $O(n_*^2)$, $O(n_*^4)$, and $O(n_*)$ choices, respectively.

For every r-graph H on V_Λ, define

$$\Phi_\Lambda(H) := \Phi(H \cap \mathcal{K}_{M_0}, M).$$

At time j, write $\hat{H}_j := H_j - \hat{B}$. Note that every cycle in this frame contains $(|V_\Lambda| - |M|)/(r-1)$ ordinary edges. Compatibility always refers to the fixed set $V(M_0)$, and the root direction orients every counted cycle uniquely.

The *directed candidates* of the frame are

$$\mathcal{R}_\Lambda := \{(P; u \to v) \colon |P| = r-2,\ u, v \notin P,\ u \ne v,\ P \cup \{u,v\} \subseteq V_\Lambda \setminus (V(M_0) \cup V(M))\}.$$

For such a candidate, let $\Psi_{\hat{H}_j, M}(P; u \to v)$ count cycles in the allowed host after deleting P and adding the pair uv, whose direction agrees with $u \to v$ relative to the frame's directed root. The candidate and its completion count are defined whether or not $P \cup \{u,v\} \in \hat{H}_j$. When this edge is present, expansion identifies the completion count with its number of cycles using the oriented edge in the frame. Moreover,

$$|\mathcal{R}_\Lambda| = r(r-1)\binom{|V_\Lambda \setminus (V(M_0) \cup V(M))|}{r} = (1+o(1))r(r-1)\binom{|V_\Lambda|}{r}.$$

Fix a sufficiently large constant B_r. The lower bound on the number of mixed Hamilton cycles that we need is

$$\log \Phi_\Lambda(\hat{H}_j) \ge \frac{|V_\Lambda| - |M|}{r-1} \log((r-1)d(\hat{H}_j)) - (|V_\Lambda| - |M|) - \frac{B_r n_*}{\sqrt{\log n_*}}. \tag{9.1}$$

For $t > 0$, $\Phi_\Lambda(\hat{H}_j) > 0$, and $d(\hat{H}_j) > 0$, a candidate is *t-abnormal* if

$$\left| \frac{(r-1)^2 d(\hat{H}_j) \Psi_{\hat{H}_j, M}(P; u \to v)}{\Phi_\Lambda(\hat{H}_j)} - 1 \right| > t.$$

The frame is *candidate-bad* if more than a $(\log\log\log n_*)^{-1/2}$ fraction of its candidates are abnormal at tolerance $(\log\log\log n_*)^{-1/2}$, and *candidate-balanced* otherwise.

Every auxiliary set T is sampled from all edges of $\hat{H}_j$. The forward edge universe is $\binom{V_\Lambda}{r}$, and the reverse universe consists of its edges missing from the remainder. Every frame cycle uses $\frac{|V_\Lambda|-|M|}{r-1}$ ordinary edges, and every completion uses $\frac{|V_\Lambda|-|M|}{r-1} - 1$. Edges excluded by the M_0-compatibility restriction remain eligible for T.

Proposition 9.3. *Fix a frame Λ and a time $m_* \le j \le \binom{n_*}{r}$. Let $\mathsf{E}_{\Lambda,j}$ be the event that the following statements hold.*

(i) F_* satisfies Proposition 5.4(i)–(ii). At the current size j, all degree and edge-distribution bounds in Proposition 8.7, with $h = 4r$ and $L = 2$, hold for H_j and its indicated vertex deletions.

(ii) The number $\Phi_\Lambda(\hat{H}_j)$ satisfies (9.1).

(iii) More than a $(\log\log\log n_*)^{-1/2}$ fraction of $\mathcal{R}_\Lambda$ are $(\log\log\log n_*)^{-1/2}$-abnormal in $\hat{H}_j$.

There is a constant $c_r > 0$ such that

$$\mathbb{P}^*[\mathsf{E}_{\Lambda,j}] \le \exp\{-c_r(\log n_*)\sqrt{\log\log\log n_*}\}. \tag{9.2}$$

The same bound holds conditionally on any $\mathbb{P}^*$-admissible values of $F_*(\hat{B}), H_j(\hat{B})$.

Proof. If too many candidates are abnormal before deletion, then most of them remain abnormal after a small random set of edges is deleted. Few of the deleted edges themselves have an abnormal orientation in the remaining hypergraph. We will show that the remaining hypergraph therefore has many abnormal missing edges, although the deleted set contains unusually few of them. Looking at the same experiment in reverse makes this discrepancy unlikely.

Fix $\mathbb{P}^*$-admissible values of $F_*(\hat{B}), H_j(\hat{B})$. If these values admit no realization in $\mathsf{E}_{\Lambda,j}$, then the required conditional probability is zero. Otherwise Lemma 8.1(i)–(ii) gives the following conditional distribution.

- The hypergraph $\hat{F}_* := F_* - \hat{B}$ is uniform among the r-graphs on V_Λ with $m_* - |F_*(\hat{B})|$ edges and the adjusted degree lower bounds ℓ'_v.
- Given $\hat{F}_*$, the hypergraph $\hat{H}_j$ is uniform among its extensions on V_Λ with $m := j - |H_j(\hat{B})|$ edges.

Put $\mathcal{E}_\Lambda := \binom{V_\Lambda}{r}$, $h := (|V_\Lambda| - |M|)/(r-1)$. These quantities are fixed by the frame and the exposed edge sets. Since these edge sets admit a realization in $\mathsf{E}_{\Lambda,j}$, we have $h \asymp_r n_*$, $m = j(1 - O_r(1/n_*))$, and $d(\hat{H}_j) \ge c_r \log n_*$.

We use $\widehat{\mathbb{P}}$ for the joint law obtained by first sampling $(\hat{F}_*, \hat{H}_j)$ according to the conditional distribution above and then choosing T uniformly among the subsets of $\hat{H}_j$ of size $\lfloor \frac{\log\log\log n_*}{100} m/h \rfloor$. Thus T is sampled from all edges of $\hat{H}_j$, and the remaining hypergraphs are $\hat{H}_j - T$ and $\hat{F}_* - T$.

For a possible remaining hypergraph with $\Phi_\Lambda(\hat{H}_j - T) > 0$, define

$$\mathcal{B}(\hat{H}_j - T) := \big\{ e \in \mathcal{E}_\Lambda \setminus (\hat{H}_j - T) : \text{some legal orientation of } e \text{ is } \frac{1}{4\sqrt{\log\log\log n_*}}\text{-abnormal in } \hat{H}_j - T \big\}.$$

Set $\mathcal{B}(\hat{H}_j - T) := \varnothing$ when $\Phi_\Lambda(\hat{H}_j - T) = 0$. This set is determined by the remaining hypergraph and the frame Λ. Let W be the event that the following statements hold.

(W1) The hypergraph $\hat{F}_* - T$ satisfies $d_{\hat{F}_* - T}(v) \ge \ell'_v$ for every $v \in V_\Lambda$.

(W2) We have $|\mathcal{B}(\hat{H}_j - T)| \ge (\log\log\log n_*)^{-1/2} |\mathcal{E}_\Lambda \setminus (\hat{H}_j - T)|/8$.

(W3) We have $|T \cap \mathcal{B}(\hat{H}_j - T)| \le (\log\log\log n_*)^{-1} |T|$.

The last two conditions express the sampling discrepancy described above. Condition (W1) ensures that every reverse extension satisfies the degree lower bounds, which will allow us to use the exact uniform law in Lemma 8.3.

We first keep a realization $(\hat{F}_*, \hat{H}_j)$ in $\mathsf{E}_{\Lambda,j}$ fixed and take probability only over T.

We next compute the probabilities that a cycle in the frame and a cycle counted by $\Psi_{\hat{H}_j, M}(P; u \to v)$ avoid T. Keep the realization fixed and use m, h defined above. The following calculations apply when $1 \le |T| \le m/4$ and $h \le m/4$. Note that the remaining host is $\hat{H}_j - T$, with $d(\hat{H}_j - T) = r(m - |T|)/|V_\Lambda|$. Here ζ_h denotes the avoidance probability in Lemma 9.1, with $t = |T|$. We have $d(\hat{H}_j - T)/d(\hat{H}_j) = (m - |T|)/m$ and

$$\frac{\zeta_{h-1}}{\zeta_h} = \frac{m - h + 1}{m - h - |T| + 1}.$$

Conditional on a present candidate edge e belonging to T, its completions avoid e. Their survival probability and its ratio to the main survival probability are

$$\widehat{\zeta}_{h-1} := \frac{\binom{m-h}{|T|-1}}{\binom{m-1}{|T|-1}}, \qquad \frac{\widehat{\zeta}_{h-1}}{\zeta_h} = \frac{m}{m - h - |T| + 1}.$$

Including the average degree, the two normalized corrections are exactly

$$\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)} \frac{\zeta_{h-1}}{\zeta_h} = 1 + \frac{|T|(h-1)}{m(m-h-|T|+1)}, \tag{9.3}$$

$$\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)} \frac{\widehat{\zeta}_{h-1}}{\zeta_h} = 1 + \frac{h-1}{m-h-|T|+1}. \tag{9.4}$$

For the prescribed size of T, $|T| = \lfloor m \log\log\log n_* / (100h) \rfloor$, these corrections are $1+o((\log\log\log n_*)^{-1/2})$ whenever the process bounds in Proposition 8.7 hold.

Put $\varepsilon := (\log\log\log n_*)^{-1/2}/10^5$. For these forward estimates, call a candidate *light* if

$$\Psi_{\hat{H}_j, M}(P; u \to v) < \Phi_\Lambda(\hat{H}_j) n_*^{-100r}.$$

Claim 9.4. *There is a bound $p_{n_*} = O_r(\varepsilon^{-2}(\log\log n_*)^{-24/25})$, uniform over the fixed realizations above, with the following properties.*

(i) Except with probability at most p_{n_*},

$$\Phi_\Lambda(\hat{H}_j - T) = (1 \pm \varepsilon)\zeta_h \Phi_\Lambda(\hat{H}_j). \tag{9.5}$$

(ii) For each candidate that is not light, except with probability at most p_{n_*},

$$\Psi_{\hat{H}_j - T, M}(P; u \to v) = (1 \pm \varepsilon)\zeta_{h-1} \Psi_{\hat{H}_j, M}(P; u \to v).$$

(iii) If such a candidate has underlying edge $e := P \cup \{u, v\} \in \hat{H}_j$, then, conditional on $e \in T$, except with probability at most p_{n_*},

$$\Psi_{\hat{H}_j - T, M}(P; u \to v) = (1 \pm \varepsilon)\widehat{\zeta}_{h-1} \Psi_{\hat{H}_j, M}(P; u \to v).$$

Proof. For a candidate that is not light, (9.1) gives

$$\log \Psi_{\hat{H}_j, M}(P; u \to v) \geq \log \Phi_\Lambda(\hat{H}_j) - 100r \log n_* \geq (h-1) \log d(\hat{H}_j) - O_r(n_*).$$

Its cycles use $h-1$ ordinary edges on the vertex set $V_\Lambda \setminus P$ and respect $M \cup \{uv\}$, with a specified relative direction. Deleting the bounded set P from the host before excluding incompatible edges removes at most $|P|\Delta_1(\hat{H}_j) = O_r(d(\hat{H}_j))$ edges. Consequently,

$$d(\hat{H}_j - P) = d(\hat{H}_j)(1 + O_r(1/n_*)), \quad \Delta_1(\hat{H}_j - P) = O_r(d(\hat{H}_j - P)).$$

These inequalities and the preceding logarithmic lower bound verify the hypotheses of Lemma 4.6 for this actual directional family, using $\hat{H}_j - P$ as a host. For the main family, the same hypotheses follow directly from (9.1) and the degree bounds on $\hat{H}_j$. In either family, two independent uniform cycles therefore share $O_r(n_*/\log d(\hat{H}_j))$ ordinary edges in expectation.

Represent each cycle by its set of ordinary edges, with its multiplicity if necessary. Each main cycle has h such edges, and each completion has $h-1$. Since $h, |T| \leq m/4$ for large n_*, Lemma 9.1 bounds the relative variance of each survival count by

$$O_r\left(\frac{\exp\{\frac{\log\log\log n_*}{25}\}}{\log d(\hat{H}_j)}\right) = O_r((\log\log n_*)^{-24/25}).$$

Here $\exp\{\frac{\log\log\log n_*}{25}\} = (\log\log n_*)^{1/25}$ and $\log d(\hat{H}_j) \geq \log\log n_* - O_r(1)$. The means are $\zeta_h \Phi_\Lambda(\hat{H}_j)$ and $\zeta_{h-1}\Psi_{\hat{H}_j, M}(P; u \to v)$, respectively. Chebyshev's inequality proves (i)–(ii), with

$$p_{n_*} = O_r\left(\frac{\varepsilon^{-2}}{(\log\log n_*)^{24/25}}\right).$$

For (iii), every cycle counted by $\Psi_{\hat{H}_j, M}(P; u \to v)$ avoids e, since its vertex set omits P. Conditional on $e \in T$, the other deleted edges form a uniform $(|T|-1)$-subset of $\hat{H}_j \setminus \{e\}$. Apply Lemma 9.1 to this universe. The overlap bound is unchanged, the relative variance has the same bound, and the mean is $\widehat{\zeta}_{h-1}\Psi_{\hat{H}_j, M}(P; u \to v)$. This proves (iii). □

The next claim moves the abnormal candidates into the missing-edge universe used by reverse sampling.

Claim 9.5. *Under the uniform choice of T for the fixed realization $(\hat{F}_*, \hat{H}_j)$, with probability $1-o(1)$,*

$$|\mathcal{B}(\hat{H}_j - T)| \geq \frac{(\log\log\log n_*)^{-1/2}}{8}|\mathcal{E}_\Lambda \setminus (\hat{H}_j - T)|.$$

Proof. Assume first that (9.5) holds. For a candidate that is not light and satisfies the survival estimate in Claim 9.4(ii), its normalized value after deletion is its normalized value before deletion multiplied by

$$\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)}\frac{\zeta_{h-1}}{\zeta_h}(1 \pm 3\varepsilon).$$

By (9.3), this factor lies in $[1-(\log\log\log n_*)^{-1/2}/10, 1+(\log\log\log n_*)^{-1/2}/10]$ for large n_*. Thus a candidate that was $(\log\log\log n_*)^{-1/2}$-abnormal remains $1/(2\sqrt{\log\log\log n_*})$-abnormal. Indeed, for every $0 < \eta \leq 1$, we have $(1-\eta/10)(1+\eta) > 1+\eta/2$ and $(1+\eta/10)(1-\eta) < 1-\eta/2$. Apply these inequalities with $\eta = (\log\log\log n_*)^{-1/2}$.

For a light candidate, monotonicity and (9.5) give

$$\frac{(r-1)^2 d(\hat{H}_j - T)\Psi_{\hat{H}_j - T, M}(P; u \to v)}{\Phi_\Lambda(\hat{H}_j - T)} \leq \frac{(r-1)^2 d(\hat{H}_j - T) n_*^{-100r}}{(1-\varepsilon)\zeta_h} = o(1),$$

since $d(\hat{H}_j - T) = O_r(n_*^{r-1})$ and $\zeta_h^{-1} \leq \exp\{2^{\frac{\log\log\log n_*}{100}}\}$. Such a candidate also remains abnormal on the low side.

By Claim 9.4(ii), the expected number of candidates that are not light and fail their survival estimate is at most $p_{n_*}|\mathcal{R}_\Lambda|$. Markov's inequality gives

$$\widehat{\mathbb{P}}\left[\text{more than } (\log\log\log n_*)^{-1/2}|\mathcal{R}_\Lambda|/10 \text{ such estimates fail} \,\middle|\, \hat{F}_*, \hat{H}_j\right] \leq \frac{10 p_{n_*}}{(\log\log\log n_*)^{-1/2}} = o(1).$$

The fixed realization lies in $\mathsf{E}_{\Lambda,j}$, so more than $(\log\log\log n_*)^{-1/2}|\mathcal{R}_\Lambda|$ candidates are initially $(\log\log\log n_*)^{-1/2}$-abnormal. Together with Claim 9.4(i), the preceding bounds show that at least $0.9(\log\log\log n_*)^{-1/2}|\mathcal{R}_\Lambda|$ candidates are $(\log\log\log n_*)^{-1/2}/2$-abnormal after deletion, with probability $1-o(1)$.

We must subtract those whose underlying edges still belong to $\hat{H}_j - T$. To use Corollary 4.5 on this host, we check its entropy lower bound. Expanding the product defining ζ_h gives

$$\log\zeta_h = -\frac{h|T|}{m} + O_r\left(\frac{\log\log\log n_*}{100 d(\hat{H}_j)} + \frac{\frac{(\log\log\log n_*)^2}{10^4}}{n_*}\right),$$
$$h\log\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)} = -\frac{h|T|}{m} + O_r\left(\frac{\frac{(\log\log\log n_*)^2}{10^4}}{n_*}\right).$$

Consequently, (9.5) and (9.1) imply

$$\log\Phi_\Lambda(\hat{H}_j - T) \geq h\log((r-1)d(\hat{H}_j - T)) - (r-1)h - \frac{(B_r+1)n_*}{\sqrt{\log n_*}}.$$

for sufficiently large n_*. The upper degree and codegree bounds are preserved because $d(\hat{H}_j - T)/d(\hat{H}_j) = 1 - o(1)$. Each partition count and its reference value change by at most $O_r(|T|)$, and

$$|T| = O_r(\frac{\log\log\log n_*}{100} d(\hat{H}_j)) = o\left(\frac{n_* d(\hat{H}_j)}{(\log n_*)^{1/8}}\right).$$

Thus the edge-distribution bound is also preserved, after a fixed enlargement of its constant. Finally, deleting the edges that fail to be M_0-compatible removes at most $O_r(n_*^{1/10} d(\hat{H}_j - T))$ edges. This verifies every hypothesis of Corollary 4.5 for the uniform family counted by $\Phi_\Lambda(\hat{H}_j - T)$.

The corollary bounds the number of $(\log\log\log n_*)^{-1/2}/4$-abnormal legal orientations whose underlying edges belong to $\hat{H}_j - T$ by

$$O_r\left(\frac{m}{(\log\log\log n_*)^{-1/2}(\log\log n_*)^{1/4}}\right) = o((\log\log\log n_*)^{-1/2} n_*^r).$$

Subtract these from the persistent abnormal candidates. Since $|\mathcal{R}_\Lambda| = (1+o(1))r(r-1)|\mathcal{E}_\Lambda|$, at least $(\log\log\log n_*)^{-1/2} r(r-1)|\mathcal{E}_\Lambda|/8$ of the remaining candidates have missing underlying edges for all sufficiently large n_*. Each edge has at most $r(r-1)$ legal orientations. Hence

$$|\mathcal{B}(\hat{H}_j - T)| \geq \frac{(\log\log\log n_*)^{-1/2}}{8}|\mathcal{E}_\Lambda| \geq \frac{(\log\log\log n_*)^{-1/2}}{8}|\mathcal{E}_\Lambda \setminus (\hat{H}_j - T)|.$$

Using the full universe in the middle expression makes this estimate valid at every host density. □

We next control the orientations of edges that actually enter T. This requires the conditional survival estimate, rather than the unconditional estimate used in the preceding claim.

Claim 9.6. *Under the uniform choice of T for the fixed realization $(\hat{F}_*, \hat{H}_j)$, with probability $1-o(1)$, $|T \cap \mathcal{B}(\hat{H}_j - T)| \leq (\log\log\log n_*)^{-1}|T|$.*

Proof. By Corollary 4.5, the number of $(\log\log\log n_*)^{-1/2}/100$-abnormal legal orientations whose underlying edges belong to $\hat{H}_j$ is

$$O_r\left(\frac{m}{(\log\log\log n_*)^{-1/2}(\log\log n_*)^{1/4}}\right).$$

Every present edge enters T with probability $|T|/m$. Therefore the expected number of these initially abnormal orientations with underlying edge in T is

$$O_r\left(\frac{|T|}{(\log\log\log n_*)^{-1/2}(\log\log n_*)^{1/4}}\right).$$

Every other legal orientation of a present edge is initially normal at tolerance $(\log\log\log n_*)^{-1/2}/100$. Its completion family is not light, since its normalized value is at least $1-(\log\log\log n_*)^{-1/2}/100$ and $d(\hat{H}_j) = O_r(n_*^{r-1})$.

Fix one such orientation with underlying edge e. Conditional on $e \in T$, Claim 9.4(iii) bounds the probability of failure of its conditional survival estimate by p_{n_*}. There are at most $r(r-1)$ orientations per edge, so the expected number of these failures among the orientations sampled into T is at most $r(r-1)|T|p_{n_*}$.

On (9.5), an initially normal orientation whose conditional estimate succeeds has its normalized value multiplied by

$$\frac{d(\hat{H}_j - T)}{d(\hat{H}_j)}\frac{\widehat{\zeta}_{h-1}}{\zeta_h}(1 \pm 3\varepsilon).$$

By (9.4), this factor lies in $[1-(\log\log\log n_*)^{-1/2}/10, 1+(\log\log\log n_*)^{-1/2}/10]$ for large n_*. Multiplication by a number in $[1-(\log\log\log n_*)^{-1/2}/100, 1+(\log\log\log n_*)^{-1/2}/100]$ therefore leaves the normalized value in $[1-(\log\log\log n_*)^{-1/2}/4, 1+(\log\log\log n_*)^{-1/2}/4]$. In particular, the orientation is normal at tolerance $(\log\log\log n_*)^{-1/2}/4$ after deletion.

Outside the failure event in Claim 9.4(i), every edge of $T \cap \mathcal{B}(\hat{H}_j - T)$ consequently has an initially abnormal orientation or an orientation whose conditional estimate failed. Markov's inequality bounds the probability that there are more than $(\log\log\log n_*)^{-1}|T|$ such edges by

$$O_r\left(\frac{1}{(\log\log\log n_*)^{-3/2}(\log\log n_*)^{1/4}} + \frac{p_{n_*}}{(\log\log\log n_*)^{-1}}\right) = o(1).$$

Adding the $p_{n_*} = o(1)$ probability of main-count failure proves the claim. This argument uses the conditional estimates for the individual completions and the unconditional estimate for the main family only through a union bound; it does not assume that main-count concentration holds conditionally on every $e \in T$. □

The remaining forward condition is that the smaller hypergraph still satisfies its degree lower bounds.

Claim 9.7. *Under the uniform choice of T for the fixed realization $(\hat{F}_*, \hat{H}_j)$, with probability $1 - o(1)$, $d_{\hat{F}_* - T}(v) \geq \ell'_v$ for every $v \in V_\Lambda$.*

Proof. Apply Lemma 8.4 with $\mathcal{U} = \hat{H}_j$, after exposing the edges at $\hat{B}$. The fixed realization satisfies the bounds on F_* in Proposition 5.4(i)–(ii), and every edge of $\mathcal{U}$ avoids $\hat{B}$. Hence the probability that an adjusted lower bound fails is at most

$$O_r\Big(\frac{\log\log\log n_*}{100} n_*^{-3/4} \log n_*\Big) + \exp\{-\Omega_r((\log n_*)^2)\} = o(1).$$

This proves the claim. □

Combining Claims 9.5 to 9.7 gives

$$\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{F}_*, \hat{H}_j] \geq 1 - \varepsilon_{n_*}, \quad \varepsilon_{n_*} = o(1), \tag{9.6}$$

uniformly over the fixed realizations in $\mathsf{E}_{\Lambda,j}$ and over the exposed edge sets. We now examine the same joint experiment in reverse, keeping the two remaining hypergraphs fixed.

Claim 9.8. *For every $\widehat{\mathbb{P}}$-admissible pair of values of $\hat{H}_j - T, \hat{F}_* - T$,*

$$\widehat{\mathbb{P}}[\mathsf{W} \mid \hat{H}_j - T, \hat{F}_* - T] \leq \exp\{-(\log\log\log n_*)^{-1/2}|T|/64\}.$$

Proof. The exposed edge sets $F_*(\hat{B}), H_j(\hat{B})$ remain fixed. If $\hat{F}_* - T$ fails an adjusted degree lower bound, or if condition (W2) in the definition of W fails, then W is impossible under this conditioning. Otherwise Lemma 8.3 applies to the full universe $\mathcal{E}_\Lambda$ and gives the exact conditional law

$$T \mid \hat{H}_j - T, \hat{F}_* - T \sim \mathrm{Unif}\binom{\mathcal{E}_\Lambda \setminus (\hat{H}_j - T)}{|T|}.$$

In particular, conditionally on these remainders, $|T \cap \mathcal{B}(\hat{H}_j - T)|$ is hypergeometric with mean at least $(\log\log\log n_*)^{-1/2}|T|/8$. For sufficiently large n_*, its threshold $(\log\log\log n_*)^{-1}|T|$ in condition (W3) is at most half this mean. The hypergeometric lower-tail bound therefore gives

$$\widehat{\mathbb{P}}\Big[|T \cap \mathcal{B}(\hat{H}_j - T)| \leq (\log\log\log n_*)^{-1}|T| \,\Big|\, \hat{H}_j - T, \hat{F}_* - T\Big] \leq \exp\{-(\log\log\log n_*)^{-1/2}|T|/64\}.$$

Since W requires this inequality on the number sampled from $\mathcal{B}(\hat{H}_j - T)$, the same upper bound holds for its conditional probability. □

We finally average the forward and reverse estimates. The following claim makes explicit that the marginal law of $(\hat{F}_*, \hat{H}_j)$ under $\widehat{\mathbb{P}}$ is the original conditional law under $\mathbb{P}^*$.

Claim 9.9. *For the fixed exposed edge sets,*

$$(1 - \varepsilon_{n_*})\mathbb{P}^*[\mathsf{E}_{\Lambda,j} \mid F_*(\hat{B}), H_j(\hat{B})] \leq \exp\{-(\log\log\log n_*)^{-1/2}|T|/64\}.$$

Proof. By (9.6) and the construction of $\widehat{\mathbb{P}}$,

$$\begin{aligned}(1 - \varepsilon_{n_*})\mathbb{P}^*[\mathsf{E}_{\Lambda,j} \mid F_*(\hat{B}), H_j(\hat{B})] &\leq \mathbb{E}_{\widehat{\mathbb{P}}}\Big[\mathbf{1}_{\mathsf{E}_{\Lambda,j}} \widehat{\mathbb{P}}[\mathsf{W} \mid \hat{F}_*, \hat{H}_j]\Big] \\ &= \widehat{\mathbb{P}}[\mathsf{E}_{\Lambda,j} \cap \mathsf{W}] \leq \widehat{\mathbb{P}}[\mathsf{W}].\end{aligned}$$

Averaging Claim 9.8 over the remainders bounds the last probability by $\exp\{-(\log\log\log n_*)^{-1/2}|T|/64\}$ and proves the claim. □

For every choice of exposed edge sets admitting a realization in $\mathsf{E}_{\Lambda,j}$, we have $m = j(1 - O_r(1/n_*))$, $j \geq m_* \asymp_r n_* \log n_*$, and $h \asymp_r n_*$. Hence the prescribed size of T is at least $c_r(\log n_*)\log\log\log n_*$, so

$$\frac{|T|}{\sqrt{\log\log\log n_*}} \geq c_r(\log n_*)\sqrt{\log\log\log n_*}.$$

Absorbing $1 - \varepsilon_{n_*}$ into the constant in Claim 9.9 proves the conditional bound (9.2). Averaging over $F_*(\hat{B}), H_j(\hat{B})$ proves the unconditional bound. □

10 Cycle modifications

In this section, we prove Proposition 7.2 by combining the candidate estimate in Proposition 9.3 with two injective modifications of mixed Hamilton cycles. These modifications show that an unusually large number of cycles for one choice of the deleted block and added pair forces comparable numbers for many choices. The first replaces one vertex of the deleted private block, and the second changes an intersection vertex of the added prescribed pair. We assume that each cycle being modified has at least 6 ordinary edges, so the auxiliary cycles below have at least 3; every family used later has $\Theta_r(n_*)$ ordinary edges.

Fix an r-graph H on $V = [n_*]$ and the original matching M_0. We use the following two settings:

$$(D_0, M) = (\varnothing, M_0), \quad (D_0, M) = (\{y_0\}, M_0 \setminus \{y_0z_0\}), \quad y_0z_0 \in M_0, \tag{10.1}$$

where either end of an original pair may be chosen as y_0. Put $\hat{H} \coloneqq H - D_0$ and $G \coloneqq \hat{H} \cap \mathcal{K}_{M_0}$. All undirected counts use the allowed host G. Directed counts whose subscript is a host also use only edges meeting $V(M_0)$ at most once. All average degrees in the probability estimates refer to hosts before excluding incompatible edges.

In the first setting, $P \cup \{y, z\}$ is disjoint from $V(M_0)$. In the second setting, $z = z_0$ and $P \cup \{y\}$ is disjoint from $V(M_0)$. In both cases $|P| = r - 2$, the labels are distinct, and the number of cycles to be compared is $\Phi(G - P, M \cup \{yz\})$. Every remaining M_0-saturated vertex belongs to a prescribed pair in each of these cycles. The matching $M \cup \{yz\}$ is nonempty in both settings. A legal target lies outside $D_0 \cup P \cup V(M_0) \cup \{y, z\}$. We orient each of these cycles by traversing the root pair yz from y to z.

We first restore one deleted private vertex by expanding an auxiliary pair; see Figure 2.

Lemma 10.1. *Write $P = S \cup \{x\}$ with $|S| = r - 3$, and fix a legal target x'. For a split $A = R \dot\cup \{u, v\}$, where $|R| = r - 3$, require that all new labels are distinct and avoid $D_0 \cup S \cup \{x, x', y, z\} \cup V(M_0)$. Define*

$$Z_{x'}(R; \{u, v\}) \coloneqq \Phi\big(G - (S \cup \{x, x'\} \cup R), M \cup \{yz, uv\}\big), \tag{10.2}$$

where uv is unordered. Then

$$\Phi(G - (S \cup \{x'\}), M \cup \{yz\}) \geq \sum_{\substack{R, \{u,v\}\ \text{legal} \\ \{x,u,v\} \cup R \in G}} Z_{x'}(R; \{u, v\}). \tag{10.3}$$

The expansion producing this inequality is injective jointly over the summation labels.

Proof. For every edge $\{x, u, v\} \cup R \in G$, replace the pair uv in an input cycle by this edge. The restored vertices are precisely $\{x\} \cup R$, so the output omits precisely $S \cup \{x'\}$. Replacing one prescribed pair by one ordinary edge preserves a single cyclic component, and every ordinary edge is allowed. Thus the output is counted by the left side of (10.3).

In the output, x is private, so its unique incident ordinary edge is recovered. The two vertices of that edge having cycle degree 2 recover $\{u, v\}$, and its other private vertices recover R. Contracting the edge recovers the input cycle. This recovers both the summation labels and the cycle, proving joint injectivity. Since uv is unordered, no orientation factor is introduced. □

To apply Proposition 9.3 later, take the frame Λ with deleted set $D_0 \cup P$, matching $M \cup \{yz\}$, directed root $y \to z$, and host $\hat{H} - P$. For a legal split in Lemma 10.1, the candidate $(\{x'\} \cup R; u \to v)$ has the same remaining vertices and prescribed pairs as the summand in (10.2). Its count specifies one relative direction. Forgetting that restriction is injective, so

$$\Psi_{\hat{H}-P, M\cup\{yz\}}(\{x'\} \cup R; u \to v) \leq Z_{x'}(R; \{u, v\}). \tag{10.4}$$

As in Section 9, the subscript records the host before incompatible edges are excluded.

We next change the intersection vertex y while keeping the deleted block P and the other intersection vertex z fixed. Consider a cycle counted by $\Phi(G - P, M \cup \{yz\})$. Starting at y away from its pair yz, follow the cycle to the next intersection vertex outside $V(M_0)$, and call it a. The initial path has one of the following forms.

(i) In Type I, the first edge is $\{y, a\} \cup R_1$, where $|R_1| = r - 2$.

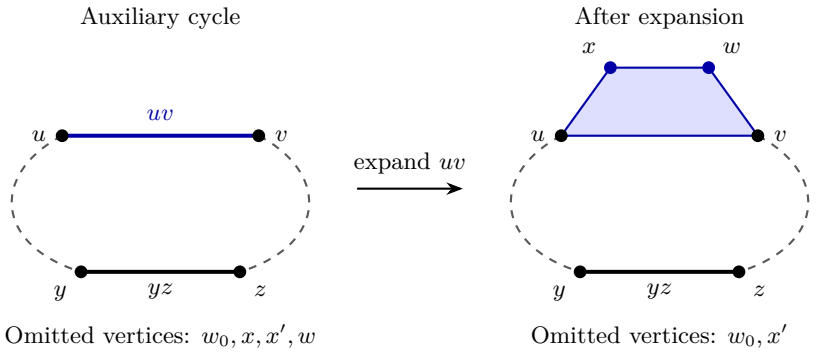


Figure 2: Private-vertex migration for $r = 4$, with $S = \{w_0\}$ and $R = \{w\}$. Expanding the prescribed pair uv into the edge $\{u, v, x, w\}$ restores x, w as private vertices. The output omits w_0, x'. Dashed arcs represent unchanged parts of the same cycle.

(ii) In Type II, the path consists of $\{y, u\} \cup R_1$, the prescribed pair $uu' \in M$, and $\{u', a\} \cup R_2$, where $|R_1| = |R_2| = r - 2$.

Indeed, the first ordinary edge reaches either an intersection vertex outside $V(M_0)$ or a M_0-saturated vertex. In the latter case the next edge is its original prescribed pair. The ordinary edge leaving the other vertex of that pair already contains a M_0-saturated vertex, so it must next reach a vertex outside $V(M_0)$. Returning to z or a previously visited intersection vertex would close a component of bounded size, contrary to the cycle being spanning and having at least 6 ordinary edges. All private labels and a lie outside $V(M_0)$.

Let $\widehat{\mathfrak{B}}(P; y, z)$ be the deterministic set of legal labeled cuts of these two types. A cut specifies the displayed vertices and private blocks, with the required distinctness and compatibility of prescribed pairs, without testing whether any ordinary edge is present. For every cut label γ, define the host before excluding incompatible edges

$$H_\gamma := \begin{cases} \hat{H} - (P \cup \{y\} \cup R_1), & \text{Type I,} \\ \hat{H} - (P \cup \{y, u, u'\} \cup R_1 \cup R_2), & \text{Type II,} \end{cases}$$

and the matching

$$M_\gamma := \begin{cases} M \cup \{az\}, & \text{Type I,} \\ (M \setminus \{uu'\}) \cup \{az\}, & \text{Type II.} \end{cases}$$

Use $a \to z$ as the directed root of the contracted cycle. Both $\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)$ and the average degree $r|H_\gamma|/|V(H_\gamma)|$ of H_γ before incompatible edges are excluded are determined after deleting y, and thus without revealing $L_{\hat{H}}(y)$.

Let P_γ be the prescribed path from y to a, and let

$$\mathfrak{B}_G(P; y, z) := \{\gamma \in \widehat{\mathfrak{B}}(P; y, z) \colon E_r(P_\gamma) \subseteq G\}$$

be the present cuts. For a present cut, contracting P_γ together with yz into the pair az is a bijection between the cycles with that initial path and the cycles counted by $\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)$. Its inverse expands az into the fixed path and pair. Each cycle counted by $\Phi(G - P, M \cup \{yz\})$ determines this initial path uniquely, so

$$\Phi(G - P, M \cup \{yz\}) = \sum_{\gamma \in \mathfrak{B}_G(P;y,z)} \Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma). \tag{10.5}$$

Only present cuts enter this sum, but $\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)$ remains the number of cycles in $H_\gamma \cap \mathcal{K}_{M_0}$ through M_γ for every fixed choice of cut labels, even if a prescribed cut edge is absent. Multiplying it by a cut-presence indicator would make it depend on the root link at y.

The number of present cuts satisfies

$$|\mathfrak{B}_G(P; y, z)| \le (r-1)d_{\hat{H}}(y) + (r-1)^2 d_{\hat{H}}(y)\Delta_1(\hat{H}) = O_r(n_*^{2r-2}). \tag{10.6}$$

For Type I, the first edge and its other intersection vertex determine R_1. For Type II, they also determine the partner u' in the prescribed pair; a second edge and its other intersection vertex then determine R_2.

The number of all possible cut labels, including labels of absent paths, is also polynomial in n_*; these choices are counted in the proof of Lemma 10.3.

We now splice a completion into a present cut. The specified relative direction makes the result connected.

Lemma 10.2. *Fix a legal target x'. For a present cut γ, let a split $A = Q \dot\cup \{v\}$, $|Q| = r-2$, be legal when $(Q; v \to x')$ is a directed candidate in its cut frame and $\{y,v\} \cup Q \in G$. Its new labels avoid the deleted cut vertices, $V(M_0)$, and the vertices of M_γ. Then*

$$\Phi(G-P, M \cup \{x'z\}) \geq \sum_{\gamma \in \mathfrak{B}_G(P;y,z)} \sum_{\substack{Q,v \text{ legal for } \gamma,x' \\ \{y,v\}\cup Q \in G}} \Psi_{H_\gamma, M_\gamma}(Q; v \to x'). \tag{10.7}$$

The map producing this inequality is injective jointly over the cuts and new-edge labels.

Proof. With directed root $a \to z$, an input cycle counted by $\Psi_{H_\gamma,M_\gamma}(Q; v \to x')$ has cyclic order

$$a \longrightarrow z \xrightarrow{A_1} v \longrightarrow x' \xrightarrow{A_2} a.$$

Remove the pairs az, vx', restore P_γ from y to a, insert $\{y,v\} \cup Q$, and add the pair $x'z$. The output has order

$$x' \longrightarrow z \xrightarrow{A_1} v \longrightarrow y \xrightarrow{P_\gamma} a \xrightarrow{A_2^{-1}} x'.$$

This is a single cyclic component. Reversing the second arc changes only its traversal, so all its undirected edges remain unchanged. The new edge restores the vertices of Q, the path restores the cut vertices, and precisely P remains absent. All ordinary edges are allowed, so the output is counted by the left side of (10.7).

To recover an input, orient the output by $x' \to z$. The edge entering y recovers v as its other intersection vertex and Q as its private block. Follow the edge leaving y to the first intersection vertex outside $V(M_0)$. This recovers the Type I or Type II path, including all cut labels. Delete the recovered ordinary cut edges, its possible internal original pair, the new ordinary edge, and $x'z$. Remove the recovered cut vertices and Q, then insert az and vx'. The remaining undirected edges are unchanged; reading the second arc in its original direction recovers the unique input. This proves injectivity jointly over cuts, new-edge labels, and input cycles. □

The direction prescribed in Lemma 10.2 ensures that the splice produces one cycle. The proof does not restrict the directions of the other prescribed pairs, so reversing the second arc preserves the required conditions. A family obtained by deleting P omits $r-2$ vertices, a Type II cut deletes a further $2r-1$, and a directed completion deletes another $r-2$. Thus at most $4r-5$ vertices are deleted from H in the first setting and $4r-4$ in the second. At most 2 original pairs are removed and at most 2 new pairs inserted. Consequently, every frame used below satisfies the bounds in Definition 9.2. After each modification, the resulting family again deletes only P from G and adds one pair to M. The additional deletions used to construct a modification do not accumulate when the operation is repeated.

We now apply the deterministic modifications to the conditioned edge-deletion process. The estimates in Proposition 9.3 concern most choices of the deleted block and added pair. To use the modifications, we also need sufficiently many suitable edges in the actual links of the vertices being restored. We specify the excluded link elements using the hypergraph with that vertex deleted, and then apply Lemma 8.6 simultaneously for the fixed choices of labels used below.

Fix $c \in (0,1]$ such that $d_{H_j}(v) \geq c\frac{rj}{n_*}$ on the event on which all conclusions of Proposition 8.7 hold. Define the constants

$$c_{\text{test}} := \frac{1}{2(r-1)^2}, \quad c_{\text{priv}} := \frac{c\,c_{\text{test}}}{6}, \quad c_{\text{int}} := \frac{c\,c_{\text{test}}}{12}, \quad c_0 := \frac{1}{2}c_{\text{priv}}^{r-2}c_{\text{int}}.$$

These are fixed before sampling. The private-vertex and intersection-vertex migrations will preserve at least the fractions c_{priv} and c_{int} of the original number of cycles. The smaller constant c_0 is used when the modified families are summed over the link of a M_0-saturated vertex.

At time j, write $H = H_j$ and fix (D_0, M) from (10.1), with $\hat H = H - D_0$ and $G = \hat H \cap \mathcal{K}_{M_0}$. Lift a residual bad set Γ of link elements at $x \notin D_0$ to the ambient link universe by

$$\widehat\Gamma := \{A \in \tbinom{V\setminus\{x\}}{r-1} \colon A \cap D_0 \neq \varnothing\} \cup \Gamma. \tag{10.8}$$

Residual link elements are regarded as subsets of $V \setminus \{x\}$. A union bound for meeting D_0 gives

$$\frac{|\widehat{\Gamma}|}{\binom{n_*-1}{r-1}} \le \frac{|\Gamma|}{\binom{n_*-|D_0|-1}{r-1}} + \frac{|D_0|(r-1)}{n_*-1}.$$

We apply every test to the ambient root link at degree $d_H(x)$. All excluded elements are included in its bad set, so every good element avoids D_0. Split legality depends only on labels and prescribed pairs; it does not test the presence of the prospective edge through the root.

For the private-vertex test, fix legal S, x, y, z, x'. Put $P = S \cup \{x\}$ and $W \coloneqq \Phi(G - P, M \cup \{yz\})$. The relevant average degree is $d(\hat{H} - P)$. Exclude residual link elements A meeting $S \cup \{x', y, z\} \cup V(M_0)$. Any other A is bad when every split $A = R \dot\cup \{u, v\}$, $|R| = r - 3$, satisfies

$$Z_{x'}(R; \{u, v\}) < \frac{c_{\mathrm{test}} W}{d(\hat{H} - P)}. \tag{10.9}$$

The hypergraphs defining W, every summand, and the average degree all omit x. Hence the excluded set in the ambient link universe is determined by $H - x$ and its fixed labels.

For the intersection-vertex test, fix all labels of $\gamma \in \widehat{\mathfrak{B}}(P; y, z)$ and a target x', without examining cut-edge presence. Exclude residual link elements at y admitting no legal split $A = Q \dot\cup \{v\}$ with candidate $(Q; v \to x')$ in the cut frame. Any other A is bad when all its legal splits satisfy

$$\Psi_{H_\gamma, M_\gamma}(Q; v \to x') < \frac{c_{\mathrm{test}} \Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)}{d(H_\gamma)}.$$

After lifting, the excluded set depends only on $H - y$ and the fixed labels. In particular, $\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)$ is defined from the hypergraph $H_\gamma \cap \mathcal{K}_{M_0}$ even when an edge of P_γ is absent.

For the test at a M_0-saturated vertex, fix an original pair $y_0 z_0$ and legal labels P, u, with $P \cup \{u\}$ disjoint from $V(M_0)$. With $G = (H - y_0) \cap \mathcal{K}_{M_0}$ and $M = M_0 \setminus \{y_0 z_0\}$, put $W \coloneqq \Phi(G - P, M \cup \{uz_0\})$. At root y_0, exclude elements meeting $V(M_0) \cup P \cup \{u\}$, and declare any other A bad when

$$\sum_{v \in A} \Phi(G - (A \setminus \{v\}), M \cup \{vz_0\}) < c_0 W.$$

The excluded set in the full ambient link universe is determined by $H - y_0$. In all three tests, invalid labels, a zero number of cycles, or a zero denominator are assigned the full bad set, which does not satisfy the density condition below.

Lemma 10.3. *For all sufficiently large n_*, the three types of link tests above require at most n_*^{4r+5} fixed choices of vertex labels. With $\mathbb{P}^*$-probability $1 - o(1)$, the following implication holds simultaneously for every such choice. If*

(i) F_* satisfies the bounds in Proposition 5.4(i)–(ii);

(ii) the process bounds in Proposition 8.7 hold; and

(iii) the excluded set in the ambient link universe has density at most $(\log\log\log n_*)^{-1/16}$,

then at least one third of the ambient root link lies outside the excluded set.

Proof. There are at most $n_* + 1$ choices of (D_0, M) and $n_*^r + 1$ times. Encode every private block by increasing order of its labels. For each such choice and time, the private-vertex tests have labels (S, x, y, z, x'), with $|S| = r - 3$, and number at most n_*^{r+1}. The Type I intersection-vertex tests have labels (P, y, z, x', a, R_1), with $|P| = |R_1| = r - 2$, and number at most n_*^{2r}. The Type II tests have labels $(P, y, z, x', u, u', a, R_1, R_2)$, with $|P| = |R_i| = r - 2$, and number at most n_*^{3r}. There are at most n_*^{r+1} labels (P, y_0, z_0, u) for the tests at M_0-saturated vertices. Thus the total is at most

$$(n_*^r + 1)\big((n_* + 1)(n_*^{r+1} + n_*^{2r} + n_*^{3r}) + n_*^{r+1}\big) \le 14 n_*^{4r+1} \le n_*^{4r+5}.$$

These fixed labels include every choice used in the two modifications. Each excluded set is defined without requiring any edge through its root to be present.

For one fixed choice of labels with root x, expose $F_* - x$ and $H_j - x$. These two hypergraphs determine the degrees at x, the total unmet degree lower bound after deleting x, and the density of the excluded

set. The bounds in Proposition 5.4(i)–(ii) give a total unmet degree lower bound of at most 1 by (5.3), and Proposition 8.7(i) gives ambient root degree at least $c_r \log n_*$. Apply Lemma 8.6 with $D = \varnothing$, conditioning only on $F_* - x$ and $H_j - x$. The probability of failure together with these conditions is at most

$$(C(\log\log\log n_*)^{-1/16})^{c_r \log n_*/4} \le \exp\{-a_r(\log n_*)\log\log\log\log n_*\} = n_*^{-\omega(1)},$$

because $\log\log\log\log n_* \to \infty$. This bounds the failure event together with the stated degree conditions. The conditional distribution used in Lemma 8.6 conditions only on $F_* - x$ and $H_j - x$, which also determine the excluded set in (10.8).

Let B_i be the failure event for fixed label choice i together with these conditions. Let E_i be any later selection event for those labels, such as requiring the cut edges to be present or the associated family to contain sufficiently many cycles. Then

$$\mathbb{P}^*\left[\bigcup_i (E_i \cap B_i)\right] \le \sum_i \mathbb{P}^*[B_i] \le n_*^{4r+5} \exp\{-a_r(\log n_*)\log\log\log\log n_*\} = o(1).$$

The same event therefore covers every fixed choice of labels. In particular, the cycle families, present cuts, and target vertices used later may be selected after the whole edge-deletion sequence has been revealed. □

Fact 10.4. *There is an event* R *of* $\mathbb{P}^*$*-probability* $1 - o(1)$ *on which the following statements hold.*

(R1) F_* *satisfies Proposition 5.4(i)–(ii).*

(R2) All conclusions of Proposition 8.7, with $h = 4r$ *and* $L = 2$*, hold throughout the edge-deletion sequence.*

(R3) At every time, in each frame satisfying (9.1), at most a $(\log\log\log n_*)^{-1/2}$ *fraction of its legal directed candidates are* $(\log\log\log n_*)^{-1/2}$*-abnormal.*

(R4) For every link test defined above whose excluded set has density at most $(\log\log\log n_*)^{-1/16}$*, at least one third of the ambient root link remains outside that set.*

Proof. Intersect the regularity events with the test event in Lemma 10.3. There are polynomially many fixed frames and times. By Proposition 9.3, each frame-time failure together with the degree and edge-distribution bounds and (9.1) is bounded by (9.2), which is $n_*^{-\omega(1)}$. A union bound over these fixed frames and times proves the assertion. □

Work on R, at a time satisfying A_j, and write $H = H_j$, $\Phi := \Phi_{M_0}(H)$. For every bounded-deletion frame Λ,

$$d(H_j - \hat{B}) = d(H)(1 + O_r(1/n_*)), \qquad \frac{|V_\Lambda| - |M|}{r-1} = k + O_r(1).$$

Indeed, the maximum-degree bound shows that deleting $O_r(1)$ vertices removes $O_r(d(H))$ edges. Since $\log d(H) = O_r(\log n_*)$, the two main terms satisfy

$$\frac{|V_\Lambda| - |M|}{r-1}\log\big((r-1)d(H_j - \hat{B})\big) - (|V_\Lambda| - |M|) \\ = k\log((r-1)d(H)) - (n_* - s) + O_r(\log n_*).$$

Therefore, if $\Phi_\Lambda(H_j - \hat{B}) \ge \Phi n_*^{-100r}$, then Lemma 7.1 and A_j give

$$\begin{aligned}\log \Phi_\Lambda(H_j - \hat{B}) &\ge \log\Phi - 100r\log n_* \\ &\ge \frac{|V_\Lambda| - |M|}{r-1}\log\big((r-1)d(H_j - \hat{B})\big) - (|V_\Lambda| - |M|) \\ &\quad - \frac{n_*}{\sqrt{\log n_*}} - O_r((s+1)\log n_*).\end{aligned}$$

The last error is $o(n_*/\sqrt{\log n_*})$. Choosing the fixed constant B_r sufficiently large gives (9.1). This argument uses the number of cycles in the actual family, including any prescribed direction.

The following two lemmas apply when the family being modified has at least Φn_*^{-3r} cycles.

Lemma 10.5. *In either setting, if* $W \coloneqq \Phi(G-(S\cup\{x\}), M\cup\{yz\}) \ge \Phi n_*^{-3r}$, *then, for all but* $O_r((\log\log\log n_*)^{-1/4}n_*)$ *legal targets* x',

$$\Phi(G-(S\cup\{x'\}), M\cup\{yz\}) \ge c_{\mathrm{priv}}W. \tag{10.10}$$

The assertion holds simultaneously for both settings, every permitted choice of vertices, and every time on R.

Proof. The frame Λ deletes $D_0\cup S\cup\{x\}$, has matching $M\cup\{yz\}$, and has root $y\to z$. The number of cycles in the frame Λ is exactly W and the average degree of its host before excluding incompatible edges is $d(\hat{H}-P)$, as in (10.9). The assumed bound $W\ge\Phi n_*^{-3r}$ and the preceding comparison give (9.1) for this family. Let $\mathcal{B}$ be its abnormal directed candidates. Since $r(r-1)\binom{n_*}{r}\le n_*^r$, candidate balance gives $|\mathcal{B}|\le(\log\log\log n_*)^{-1/2}n_*^r$. A normal legal candidate satisfies, by (10.4),

$$Z_{x'}(R;\{u,v\}) \ge \Psi_{\hat{H}-P,M\cup\{yz\}}(\{x'\}\cup R; u\to v) \ge \frac{(1-(\log\log\log n_*)^{-1/2})W}{(r-1)^2 d(\hat{H}-P)} \ge \frac{c_{\mathrm{test}}W}{d(\hat{H}-P)}$$

for large n_*. Thus a low summand forces an abnormal candidate, using the average degree before excluding incompatible edges throughout.

Claim 10.6. *For all but* $O_r((\log\log\log n_*)^{-1/4}n_*)$ *legal targets, the lifted bad set of the private-vertex test has density at most* $(\log\log\log n_*)^{-1/16}$.

Proof. Put $\mathcal{B}_{x'} \coloneqq \{(Q;u\to v)\in\mathcal{B}\colon x'\in Q\}$ and $\beta_0 \coloneqq \sqrt{(r-2)(\log\log\log n_*)^{-1/2}}$. Since every candidate private block has $r-2$ vertices,

$$\sum_{x'}|\mathcal{B}_{x'}| = (r-2)|\mathcal{B}| \le \beta_0^2 n_*^r.$$

At most $\beta_0 n_*$ targets have $|\mathcal{B}_{x'}|>\beta_0 n_*^{r-1}$. Also exclude the at most $L\coloneqq 2s+r+1$ labels in $D_0\cup P\cup V(M_0)\cup\{y,z\}$.

Fix a remaining target and put $Z\coloneqq D_0\cup S\cup\{x'\}\cup V(M_0)\cup\{y,z\}$, so $|Z|\le L$. For each ambient link element A at x avoiding Z, choose a deterministic split $A=R\,\dot\cup\,\{u,v\}$ and direct uv by increasing label order. If A is bad, this split is low and gives an abnormal candidate $(\{x'\}\cup R;u\to v)$ in $\mathcal{B}_{x'}$. This assignment is injective because the candidate recovers $A=(Q\setminus\{x'\})\cup\{u,v\}$. Every other bad ambient link element meets Z. Hence

$$\begin{aligned}\frac{|\widehat{\Gamma}_{x'}|}{\binom{n_*-1}{r-1}} &\le \frac{\beta_0 n_*^{r-1}}{\binom{n_*-1}{r-1}} + \frac{L(r-1)}{n_*-1}\\ &\le 2^{r-1}(r-1)!\,\beta_0 + \frac{L(r-1)}{n_*-1} = o((\log\log\log n_*)^{-1/16}).\end{aligned} \tag{10.11}$$

We used $n_*\ge 2r$, $s\le n_*^{1/10}$, and $(\log\log\log n_*)^{-1/4}=o((\log\log\log n_*)^{-1/16})$. The density is measured in the ambient universe and includes collisions with the base deletion. There are $\beta_0 n_*+L = O_r((\log\log\log n_*)^{-1/4}n_*)$ excluded targets. □

For a target supplied by Claim 10.6, the private-vertex link estimate in Lemma 10.3 gives at least $d_H(x)/3$ good ambient root edges, all remaining in G. Choose one good unordered split from each such edge and apply the jointly injective map in Lemma 10.1. Since $d(\hat{H}-P)\le 2d(H)$ and $d_H(x)\ge cd(H)$,

$$\Phi(G-(S\cup\{x'\}), M\cup\{yz\}) \ge \frac{d_H(x)}{3}\frac{c_{\mathrm{test}}W}{d(\hat{H}-P)} \ge \frac{c\,c_{\mathrm{test}}}{6}W = c_{\mathrm{priv}}W.$$

This proves (10.10). □

Lemma 10.7. *In either setting, suppose* $W\coloneqq\Phi(G-P, M\cup\{yz\})\ge\Phi n_*^{-3r}$ *and* $y\notin V(M_0)$. *Then, for all but* $O_r((\log\log\log n_*)^{-1/8}n_*)$ *legal targets,*

$$\Phi(G-P, M\cup\{x'z\}) \ge c_{\mathrm{int}}W.$$

Proof. Partition the family counted by W using (10.5). Discard present cuts with $\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma) < Wn_*^{-(2r+2)}$. By (10.6), the discarded families contain $O_r(n_*^{-4})W$ cycles in total. Every retained cut has at least $\Phi n_*^{-(5r+2)} \geq \Phi n_*^{-100r}$ cycles and hence satisfies (9.1).

Claim 10.8. *For each retained cut γ, there is a set T_γ of $O_r((\log\log\log n_*)^{-1/4}n_*)$ targets such that every target outside T_γ is legal for the cut and its lifted intersection-vertex test has density at most $(\log\log\log n_*)^{-1/16}$.*

Proof. Candidate balance gives at most $(\log\log\log n_*)^{-1/2}n_*^r$ abnormal directed candidates in the cut frame. Sum their counts over their second directed vertex x'; each candidate is counted once. Thus at most $(\log\log\log n_*)^{-1/4}n_*$ targets have more than $(\log\log\log n_*)^{-1/4}n_*^{r-1}$ such candidates.

For a remaining target, any bad link element avoiding the excluded labels maps to an abnormal candidate by a fixed split $A = Q \dot\cup \{v\}$ and direction $v \to x'$. The map is injective because the candidate recovers $A = Q \cup \{v\}$. A normal candidate would instead give $\Psi_{H_\gamma,M_\gamma}(Q; v \to x') \geq c_{\mathrm{test}}\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)/d(H_\gamma)$. Base, cut, target, and M_0-saturated-vertex exclusions account for at most $O_r((s+1)n_*^{r-2})$ link elements. Dividing by the ambient denominator as in (10.11) gives bad density

$$O_r((\log\log\log n_*)^{-1/4}) + O_r((s+1)/n_*) = o((\log\log\log n_*)^{-1/16}).$$

Add targets colliding with the cut or its prescribed pairs to T_γ. Their number is $O_r(s+1) = o((\log\log\log n_*)^{-1/4}n_*)$, proving the claimed bound uniformly in γ. □

Claim 10.9. *Outside $O_r((\log\log\log n_*)^{-1/8}n_*)$ targets, the retained present cuts with $x' \notin T_\gamma$ have total count at least $W/2$.*

Proof. By Claim 10.8, each retained cut has $O_r((\log\log\log n_*)^{-1/4}n_*)$ exceptional targets. Weight them by the actual cut counts. Summing only over retained present cuts,

$$\begin{aligned}\sum_{x'} \sum_{\substack{\gamma \text{ retained} \\ x' \in T_\gamma}} \Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma) &= \sum_{\gamma \text{ retained}} |T_\gamma| \Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma) \\ &\leq \frac{C_r n_*}{(\log\log\log n_*)^{1/4}} \sum_{\gamma \text{ retained}} \Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma) \\ &\leq \frac{C_r n_* W}{(\log\log\log n_*)^{1/4}}.\end{aligned}$$

The preceding inequality bounds the sum of the exceptional cut counts over all targets. By Markov's inequality, all but $O_r(n_*/(\log\log\log n_*)^{1/8})$ targets have exceptional cut count at most $W/(\log\log\log n_*)^{1/8}$. The discarded cuts contribute $O_r(n_*^{-4})W$. For every other target, the two losses total at most

$$\frac{W}{(\log\log\log n_*)^{1/8}} + O_r(n_*^{-4})W = o(W).$$

The retained good cuts therefore have total count at least $W/2$. This uses a weighted sum, so different cuts may have different good target sets. □

Fix a target supplied by Claim 10.9. For every retained good cut, the intersection-vertex link estimate in Lemma 10.3 gives at least $d_H(y)/3$ good ambient link elements. They avoid the excluded labels and provide legal new edges in G. Choose one good split per element. Since $d(H_\gamma) \leq 2d(H)$, the contribution of this cut in Lemma 10.2 is at least

$$\frac{d_H(y)}{3} \frac{c_{\mathrm{test}}\Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma)}{d(H_\gamma)} \geq \frac{c\, c_{\mathrm{test}}}{6} \Phi(H_\gamma \cap \mathcal{K}_{M_0}, M_\gamma).$$

Joint injectivity over cuts and splits allows these contributions to be summed. By Claim 10.9, their sum is at least $(c\, c_{\mathrm{test}}/12)W = c_{\mathrm{int}}W$. □

Proof of Proposition 7.2. Work on R and fix a time for which A_j holds. Write $H = H_j$, $\Phi = \Phi_{M_0}(H)$. We bound marginals first for edges avoiding $V(M_0)$, then for edges containing one M_0-saturated vertex.

Claim 10.10. *Every allowed edge e avoiding $V(M_0)$ has $q_e(H) = O_r(1/d(H))$.*

Proof. Use the first setting $G = H \cap \mathcal{K}_{M_0}$, $M = M_0$, and let B_H be the maximum of $\Phi(G - P, M \cup \{yz\})$ over labels outside $V(M_0)$. If $B_H < \Phi n_*^{-2r}$, then $B_H = o(\Phi/d(H))$ because $d(H) = O_r(n_*^{r-1})$. Otherwise start at a choice attaining B_H, replace its $r-2$ private vertices one at a time using Lemma 10.5, then change its two intersection vertices using Lemma 10.7. Before the second intersection-vertex change, use $\Phi(G - P, M \cup \{yz\}) = \Phi(G - P, M \cup \{zy\})$. The number of changes is fixed, and only fixed positive factors are lost, so every intermediate family has at least Φn_*^{-3r}.

At each stage, all but $O_r((\log\log\log n_*)^{-1/8} n_*)$ choices succeed for each preceding family. There are r stages in total. For each unsuccessful stage after a successful prefix, completing the remaining coordinates arbitrarily accounts for at most $O_r((\log\log\log n_*)^{-1/8} n_*^r)$ failed sequences. Summing over the fixed number of stages has the same order. Collisions with fixed labels or $V(M_0)$ account for $O_r((s+1)n_*^{r-1}) = o((\log\log\log n_*)^{-1/8} n_*^r)$ further sequences. Every final private block has at most $(r-2)!$ ordered representations. Thus, outside $O_r((\log\log\log n_*)^{-1/8} n_*^r)$ legal labeled choices,

$$\Phi(G - P', M \cup \{y'z'\}) \geq c_{\mathrm{priv}}^{r-2} c_{\mathrm{int}}^2 B_H. \tag{10.12}$$

In the original frame, candidate balance in both directions and (2.3) give

$$\Phi(G - P', M \cup \{y'z'\}) \leq \frac{2(1 + (\log\log\log n_*)^{-1/2})\Phi}{(r-1)^2 d(H)} \tag{10.13}$$

outside $O_r((\log\log\log n_*)^{-1/2} n_*^r)$ legal choices. The legal universe has order n_*^r, and the two exceptional sets have size $o(n_*^r)$. Therefore (10.12) and (10.13) hold together for some choice, giving $B_H = O_r(\Phi/d(H))$. For an edge $e \in G$ avoiding $V(M_0)$, (2.4) gives

$$\Phi q_e(H) = \sum_{\{y,z\} \in \binom{e}{2}} \Phi\big(G - (e \setminus \{y,z\}), M \cup \{yz\}\big) \leq \binom{r}{2} B_H = O_r\left(\frac{\Phi}{d(H)}\right).$$

Dividing by Φ proves the claim. □

Claim 10.11. *Every allowed edge e containing exactly one M_0-saturated vertex has $q_e(H) = O_r(1/d(H))$.*

Proof. Fix an edge e as in the claim, let y_0 be its unique M_0-saturated vertex, and let $y_0 z_0 \in M_0$. For each $u \in e \setminus \{y_0\}$, we count the mixed Hamilton cycles in which y_0, u are the intersection vertices of e and $P = e \setminus \{y_0, u\}$ is its private block. Contracting e and the consecutive pair $y_0 z_0$ gives this number as

$$\Phi_e := \Phi(G - P, M \cup \{uz_0\}), \quad G = (H - y_0) \cap \mathcal{K}_{M_0}, \quad M = M_0 \setminus \{y_0 z_0\}.$$

This contracts a path and preserves a single cyclic component. If $\Phi_e < \Phi n_*^{-2r}$, the required bound is immediate. Otherwise replace the $r-2$ private vertices one at a time and then change the intersection vertex u, keeping z_0 fixed. These modifications stay in the second setting and give

$$\Phi(G - P', M \cup \{vz_0\}) \geq c_{\mathrm{priv}}^{r-2} c_{\mathrm{int}} \Phi_e$$

for all but $O_r((\log\log\log n_*)^{-1/8} n_*^{r-1})$ legal choices (P', v). Every family used has at least Φn_*^{-3r}.

Since $c_0 < c_{\mathrm{priv}}^{r-2} c_{\mathrm{int}}$, every split of a bad legal link element in the link test at y_0 defined above must be one of those failed choices. Choose one split deterministically; it recovers the link element as $A = P' \cup \{v\}$, so this assignment is injective. Including excluded labels, the bad density is

$$O_r((\log\log\log n_*)^{-1/8}) + O_r((s+1)/n_*) = o((\log\log\log n_*)^{-1/16}).$$

The corresponding link estimate in Lemma 10.3 consequently gives at least $d_H(y_0)/3$ good ambient link elements.

Every original mixed cycle has exactly one ordinary edge at y_0 and exactly one other intersection vertex in that edge. Partitioning the cycles by this edge and vertex gives

$$\Phi = \sum_{\substack{A \in L_H(y_0) \\ A \cap V(M_0) = \varnothing}} \sum_{v \in A} \Phi(G - (A \setminus \{v\}), M \cup \{vz_0\}).$$

Conversely, expanding a summand recovers the ordinary edge and original pair uniquely. Each good link element contributes at least $c_0\Phi_e$, and hence

$$\Phi \geq \frac{1}{3} d_H(y_0) c_0 \Phi_e, \quad \Phi_e = O_r(\Phi/d(H)).$$

Sum over the at most $r-1$ choices of u in the edge to obtain its marginal bound. □

By Claims 10.10 and 10.11, every allowed edge has marginal $O_r(1/d(H))$. Edges meeting two M_0-saturated vertices are forbidden and have marginal zero. Finally,

$$\frac{k}{j} = \frac{r(1-s/n_*)}{(r-1)d(H)},$$

so the bound is $O_r(k/j)$. It holds simultaneously at every time because R covers all fixed choices of labels and times. Since $\mathbb{P}^*[\mathsf{R}] = 1 - o(1)$, this proves Proposition 7.2. □

11 Counting at the hitting time

We prove Theorem 1.2. Put $h := n/(r-1)$ and define the reference value for the logarithm of the number of cycles

$$\lambda_r(n,m) := h \log \frac{r(r-1)m}{n} - n.$$

The upper bound follows from the enumeration in the complete hypergraph and a first-moment estimate applied simultaneously at deterministic process times.

Lemma 11.1. *For an integer $h = n/(r-1) \geq 3$, the number of loose Hamilton cycles in the complete r-graph on $[n]$ is*

$$T_{n,r} = \frac{n!}{2h((r-2)!)^h}.$$

Uniformly for $h \leq m \leq \binom{n}{r}$, if H_m is a uniform m-edge r-graph, then

$$\mathbb{E}[Z(H_m)] = T_{n,r} \frac{(m)_h}{(\binom{n}{r})_h} \leq \exp\{\lambda_r(n,m) + O_r(\log n)\}. \tag{11.1}$$

Proof. Choose the h intersection vertices and their cyclic order up to reversal, then distribute the remaining vertices into the h private blocks. This gives

$$\binom{n}{h} \frac{(h-1)!}{2} \frac{(n-h)!}{((r-2)!)^h} = \frac{n!}{2h((r-2)!)^h}.$$

Each fixed cycle is contained in H_m with probability $(m)_h/(\binom{n}{r})_h$. For $0 \leq i < h$ we have $(m-i)/(\binom{n}{r}-i) \leq m/\binom{n}{r}$, so this probability is at most $(m/\binom{n}{r})^h$. Stirling's formula and $\binom{n}{r} = n^r/r!\,(1+O_r(1/n))$ give

$$\log T_{n,r} + h \log(m/\binom{n}{r}) = h \log \frac{r(r-1)m}{n} - n + O_r(\log n),$$

using $(r-1)! = (r-1)(r-2)!$. This proves (11.1). □

Proof of Theorem 1.2. For the upper bound, apply Lemma 11.1 to every deterministic process time in

$$\mathcal{I}_n := \left[\frac{n \log n}{2r}, \frac{2n \log n}{r} \right] \cap \mathbb{Z}.$$

By Markov's inequality and a union bound, for large n,

$$\mathbb{P}[\exists m \in \mathcal{I}_n : Z(\mathcal{H}_m) > \exp\{\lambda_r(n,m) + 2n/\log n\}] \leq |\mathcal{I}_n| \exp\{-n/\log n\} = o(1).$$

By (6.2), $\tau_1 \in \mathcal{I}_n$ with high probability. The simultaneous estimate therefore gives the required upper bound at τ_1.

For the lower bound, use the exposure and core in Proposition 6.4. Since the matching expansion is injective, Theorem 6.1, with offset allowance $10r$, gives

$$\log Z(\mathcal{H}_{\tau_1}) \geq k \log \frac{r(r-1)m_*}{n_*} - (r-1)k - C_r \frac{n_*}{\log n_*}$$

with failure probability tending to zero uniformly over the admissible exposures. Write $t := |T_D| = O_r(s \log n)$, where $s \leq n^{1/12}$. We have $n_* = n - (r-2)s$ and $k = h - s$. Consequently, $k = h - O_r(s)$, $(r-1)k = n - O_r(s)$, and

$$\frac{nm_*}{n_*\tau_1} = \frac{1 - t/\tau_1}{n_*/n} = 1 + O_r(s/n).$$

Since $r\tau_1/n = \Theta(\log n)$, we obtain

$$\begin{aligned} k \log \frac{r(r-1)m_*}{n_*} - (r-1)k &= h \log \frac{r(r-1)\tau_1}{n} - n + O_r(s \log\log n) \\ &= \lambda_r(n, \tau_1) + o(n/\log n). \end{aligned}$$

Also $n_*/\log n_* = (1 + o(1))n/\log n$. Averaging the conditional lower bound over the admissible exposures and using Proposition 6.4 proves (1.1). Finally, (6.2) gives

$$\lambda_r(n, \tau_1) = \frac{n}{r-1} \log((r-1)\log n) - n + O_r\left(\frac{n \log\log n}{\log n}\right).$$

Together with (1.1), this proves (1.2). □

12 Concluding remarks

The proof applies to each fixed $r \geq 4$. At $r = 3$, a private block consists of one vertex, and the private-vertex migration already changes the whole block. For higher uniformity, the same comparison is applied successively to its $r - 2$ coordinates. The number of changes, auxiliary vertices, and link conditions remains bounded in terms of r.

A Lean 4 formalization of the corresponding hitting-time statement for $r \geq 3$ is available at the source revision cited in León [24] and registered as Palomar entry `PALOMAR-2026-10-03-000003`, version 1. The quantitative conclusion in Theorem 1.2 is proved here by the conditioned-core lower bound, the injective expansion, and the first-moment estimate applied simultaneously at deterministic process times.

Declaration on the use of AI

GPT 6 Pro was used to assist in organizing the proof, checking calculations, and improving the exposition of this draft.

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